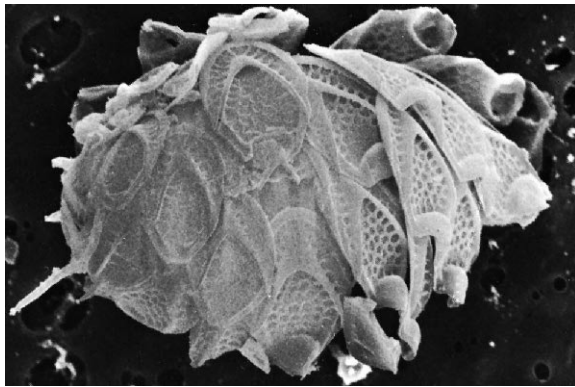
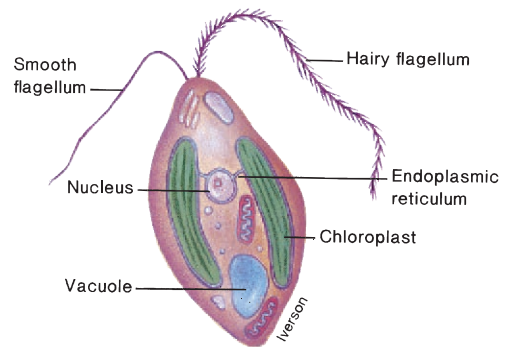
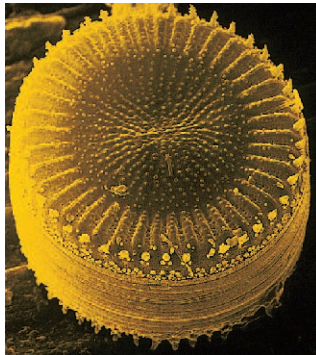




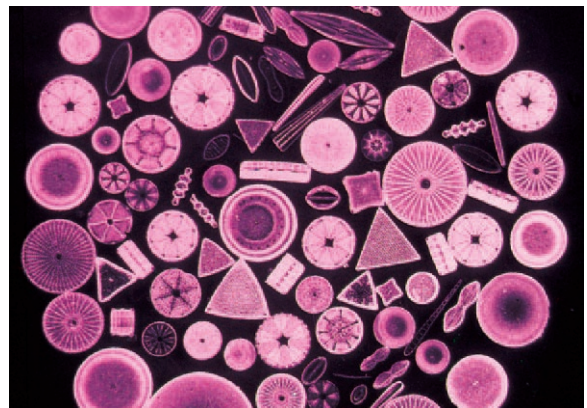
(a)



(b)



(c)



(d)

Figure 26.6 Chrysophyta (Yellow-Green and Golden-Brown Algae; Diatoms). (a) Scanning electron micrograph of *Mallomonas*, a chrysophyte, showing its silica scales. The scales are embedded in the pectin wall but synthesized within the Golgi apparatus and transported to the cell surface in vesicles ($\times 9,000$). (b) *Ochromonas*, a unicellular chrysophyte. Diagram showing typical cell structure. (c) Scanning electron micrograph of a diatom, *Cyclotella meneghiniana* ($\times 750$). (d) Assorted diatoms as arranged by a light microscope ($\times 900$).

the canal and actively beats to move the cell. Reproduction in euglenoids is by longitudinal mitotic cell division.

Chrysophyta (Golden-Brown and Yellow-Green Algae; Diatoms)

The division *Chrysophyta* is quite diversified with respect to pigment composition, cell wall, and type of flagellated cells. In molecular classification schemes, these algae are associated with the stramenopiles and have mitochondria with tubular cristae (Phylogenetic Diagram 26). The division is divided into three major classes: golden-brown algae [Greek *chrysos*, gold], yellow-green algae, and diatoms. The major photosynthetic pigments are usually chlorophylls *a* and *c₁/c₂*, and the carotenoid fucoxanthin. When fucoxanthin is the dominant pigment, the cells have a golden-brown color. The major carbohydrate reserve in the *Chrysophyta* is **chrysolaminarin** (a polysaccharide storage product composed principally of β -1,3 linked glucose residues).

Some *Chrysophyta* lack cell walls; others have intricately patterned coverings external to the plasma membrane, such as **scales** (figure 26.6a), walls, and plates. Diatoms have a distinctive two-piece wall of silica, called a **frustule**. Two anteriorly attached flagella of unequal length are common among *Chrysophyta* (figure 26.6b), but some species have no flagella, and others have either one flagellum or two that are of equal length.

Most *Chrysophyta* are unicellular or colonial. Reproduction usually is asexual but occasionally sexual. Although some marine forms are known, most of the yellow-green and golden-brown algae live in fresh water. Blooms of some species produce unpleasant odors and tastes in drinking water.

The **diatoms** (figure 26.6c,d; see also figure 4.1b) are photosynthetic, circular or oblong chrysophyte cells with frustules composed of two halves or thecae that overlap like a petri dish [therefore their name is from the Greek *diatomsos*, cut in two]. The larger half is the **epitheca**, and the smaller half is the **hypotheca**. Diatoms grow in freshwater, salt water, and moist soil and comprise a large

Box 26.1

Practical Importance of Diatoms

Diatoms have both direct and indirect economic significance for humans. Because diatoms make up most of the phytoplankton of the cooler parts of the ocean, they are the most important ultimate source of food for fish and other marine animals in these regions. It is not unusual for 1 liter of seawater to contain almost a million diatoms.

When diatoms die, their frustules sink to the bottom. Because the siliceous part of the frustule is not affected by the death of the cell, diatom frustules tend to accumulate at the bottom of aquatic environments. These form deposits of material called **diatomaceous earth**. This material is used as an active ingredient in many commercial preparations, including detergents, fine abrasive polishes, paint removers, decolorizing and deodorizing oils, and fertilizers. Diatomaceous earth also is used extensively as a filtering agent, as a component in insulating (firebrick) and

soundproofing products, and as an additive to paint to increase the night visibility of signs and license plates.

The use of diatoms as indicators of water quality and of pollution tolerance is becoming increasingly important. Specific tolerances for given species to various environmental parameters (concentrations of salts, pH, nutrients, nitrogen, temperature) have been compiled.

Recently a diatomaceous earth product to control insects, called INSECTO, has been introduced. Insects have their soft body parts exposed but covered by a waxy film to prevent dehydration. When they contact the diatoms in INSECTO, the silica frustules break the waxy film on the insects, causing them to dehydrate and die. INSECTO is a physical control product to which insects cannot build up resistance. Yet this product can be fed to poultry, livestock, and pets with no ill effects.

part of the phytoplankton (**Box 26.1**). The chloroplasts of these chrysophytes contain chlorophylls *a* and *c* as well as carotenoids. Some diatoms are facultative heterotrophs and can absorb carbon-containing molecules through the holes in their walls. The vegetative cells of diatoms are diploid; exist as unicellular, colonial, or filamentous shapes; lack flagella; and have a single large nucleus and smaller plastids. Reproduction consists of the organism dividing asexually, with each half then constructing a new theca within the old one. Because of this mode of reproduction, diatoms get smaller with each reproductive cycle. However, when they diminish to about 30% of their original size, sexual reproduction usually occurs. The diploid vegetative cells undergo meiosis to form gametes, which then fuse to produce a zygote. The zygote develops into an auxospore, which increases in size again and forms a new wall. The mature auxospore eventually divides mitotically to produce vegetative cells with normal frustules.

Diatom frustules are composed of crystallized silica [Si(OH)₄] with very fine markings (figure 26.6c,d). They have distinctive, and often exceptionally beautiful, patterns that are different for each species. Frustule morphology is very useful in diatom identification.

1. Why are the *Charophyta* called stoneworts?
2. Briefly describe the major characteristics of the euglenoids, *Chrysophyta*, and diatoms.
3. How do euglenoids and diatoms reproduce?
4. How do the cells of diatoms differ from those of other organisms?

Phaeophyta (Brown Algae)

The *Phaeophyta* or brown algae [Greek *phaeo*, brown] consist of multicellular organisms that occur almost exclusively in the sea. Some species have the largest linear dimensions (length) known

in the eucaryotic world (chapter opening figure). Since the brown algae have tubular cristae, they are associated with stramenopiles in molecular classification schemes (Phylogenetic Diagram 26). Most of the conspicuous seaweeds that are brown to olive green in color are assigned to this division. The simplest brown algae consist of small openly branched filaments; the larger, more advanced species have a complex arrangement. Some large **kelps** are conspicuously differentiated into flattened blades, stalks, and holdfast organs that anchor them to rocks (**figure 26.7**). Some, such as *Sargassum*, form huge floating masses that dominate the Sargasso Sea. The color of these algae reflects the presence of the brown pigment fucoxanthin, in addition to chlorophylls *a* and *c*, β -carotene, and violaxanthin. The main storage product is **laminarin**, which is quite similar in structure to chrysolaminarin.

Rhodophyta (Red Algae)

The division Rhodophyta, the red algae [Greek *rhodon*, rose], includes most of the seaweeds (**figure 26.8**). A few reds are unicellular but most are filamentous and multicellular. Some red algae are up to 1 m long. The stored food is the carbohydrate called floridean starch (composed of α -1,4 and α -1,6 linked glucose residues).

The red algae contain the red pigment phycoerythrin, one of the two types of phycobilins that they possess. The other accessory pigment is the blue pigment phycocyanin. The presence of these pigments explains how the red algae can live at depths of 100 m or more. The wavelengths of light (green, violet, and blue) that penetrate these depths are not absorbed by chlorophyll *a* but instead by these phycobilins. Not surprisingly the concentrations of these pigments often increases with depth as light intensity decreases. The phycobilins, after absorbing the light energy, pass it on to chlorophyll *a*. The algae appear decidedly red when phycoerythrin predominates over the other pigments. When phycoerythrin un-

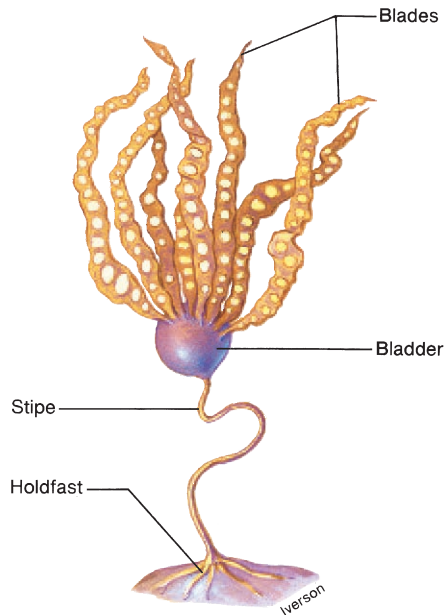


Figure 26.7 Phaeophyta (Brown Algae). Diagram of the parts of the brown alga, *Nereocystis*. Due to the holdfast organ, the heaviest tidal action and surf seldom dislodge brown algae from their substratum. The stipe is a stalk that varies in length; the bladder is a gas-filled float.

dergoes photodestruction in bright light, other pigments predominate and the algae take on shades of blue, brown, and dark green.

The cell walls of most red algae include a rigid inner part composed of microfibrils and a mucilaginous matrix. The matrix is composed of sulfated polymers of galactose called agar, funori, porphyran, and carrageenan. These four polymers give the red algae their flexible, slippery texture. Agar is used extensively in the laboratory as a culture medium component (see section 5.7). Many red algae also deposit calcium carbonate in their cell walls and play an important role in building coral reefs.

***Pyrrhophyta* (Dinoflagellates)**

The *Pyrrhophyta* or **dinoflagellates** are unicellular, photosynthetic alveolate algae (Phylogenetic Diagram 26). Most dinoflagellates are marine, but some live in fresh water. Along with the chrysophytes and diatoms, the dinoflagellates make up a large part of freshwater and marine plankton and are at the base of many food chains. Species of *Noctiluca*, *Pyrodinium*, *Gonyaulax*, and other genera can produce light and are responsible for much of the luminescence (phosphorescence) seen in ocean waters at night. Sometimes dinoflagellate populations reach such high levels that poisonous red tides result (**Box 26.2**).



Figure 26.8 Rhodophyta (Red Algae). These algae (e.g., *Corallina gracilis*) are much smaller and more delicate than the brown algae. Most red algae have a filamentous, branched morphology as seen here.

The flagella, protective coats (plates), and biochemistry of the dinoflagellates are distinctive. Many dinoflagellates are armored or clad in stiff, patterned, cellulose plates or thecae, which may become encrusted with silica (**figure 26.9**). Most have two flagella. In armored dinoflagellates these flagella beat in two grooves that girdle the cell—one transverse (the cingulum), the other longitudinal (the sulcus). The longitudinal flagellum extends posteriorly as a rudder; the flattened, ribbonlike transverse flagellum propels the cell forward while causing it to spin. Due to this spinning, the dinoflagellates received their name from the Greek *dinein*, “to whirl.”

Most dinoflagellates have chlorophylls *a* and *c*, in addition to carotenoids and xanthophylls. As a result they usually have a yellowish-green to brown color. Their mitochondria have tubular cristae.

Some dinoflagellates can ingest other cells; others are colorless and heterotrophic. A few even occur as symbionts in many groups of jellyfish, sea anemones, mollusks, and corals. When dinoflagellates form symbiotic relationships, they lose their cellulose plates and flagella, become spherical golden-brown globules in the host cells, and are then termed **zooxanthellae**.

1. What is a seaweed?
2. Describe a kelp.
3. Why do red algae appear red?
4. What is unique about the flagellar arrangement in the dinoflagellates?

Box 26.2

Toxic Algal Blooms

The Bible reports that the first plague Moses visited on the Egyptians was a blood-red tide that killed fish and fouled water. The Red Sea probably is named after these toxic algal blooms. Thousands of years later we still have problems with this plague.

The poisonous and destructive **red tides** that occur frequently in coastal areas often are associated with population explosions, or “blooms,” of dinoflagellates. *Gymnodinium* and *Gonyaulax* species are the dinoflagellates most often involved. The pigments in the dinoflagellate cells are responsible for the red color of the water. Under these bloom conditions, the dinoflagellates produce a powerful neurotoxin called saxitoxin. The toxin paralyzes the striated respiratory muscles in many vertebrates by inhibiting sodium transport, which is essential to the function of their nerve cells. The toxin does not harm the shellfish that feed on the dinoflagellates. However, the shellfish do accumulate the toxin and are themselves highly poisonous to organisms, such as humans, who consume the shellfish, resulting in a condition known as **paralytic shellfish poisoning** or neurotoxic shellfish poisoning. Paralytic shellfish poisoning is characterized by numbness of the mouth, lips, face, and extremities. Duration of the illness ranges from a few hours to a few days and usually is not fatal.

Another type of poisoning in humans is called **ciguatera**. It results from eating marine fishes (e.g., grouper, snapper) that have consumed the dinoflagellate *Gambierdiscus toxicus*. The alga’s toxin, called ciguatoxin, accumulates in the flesh of fish. This is one of the most powerful toxins known and remains in the flesh even after it has been cooked. Unfortunately it cannot be detected in the fishes and they are not visibly affected. In humans the toxin may cause gastrointestinal disturbances, profuse diarrhea, central nervous system involvement, and respiratory failure.

In 1988 a red tide that has long plagued the Gulf Coast of Florida spread northward to North Carolina. The dinoflagellates released a neu-

rotoxin called brevetoxin, and this prompted state health authorities to shut down all shellfishing for three months. In 1987, in Prince Edward Island, Canada, several people died and hundreds became sick from eating mussels contaminated with domoic acid. The domoic acid was traced to a bloom of diatoms, golden-brown algae once thought to be innocent of all toxicity. The resulting disease, called **amnesic shellfish poisoning**, produces short-term memory loss in its victims. In 1991 pelicans eating anchovies off the California coast were found to be dying from domoic acid poisoning. Shellfish and crab fisheries from Washington to California were closed for several months, resulting in losses reaching hundreds of millions of dollars. In 1993 saxitoxin was found for the first time in crabs from Alaska. Unfortunately there are no treatments for the above types of poisonings. Supportive measures are the only therapy.

Overall, toxic algal blooms are on the rise. For example, in 1997 *Pfiesteria piscicida* (Latin for “fish killer”) and other *Pfiesteria*-like dinoflagellates caused large fish kills along the coast of Maryland and Virginia. Similar fish kills have been occurring along the Atlantic coast at least since the 1980s. The flagellated form of the ambush-predator dinoflagellate swims toward the fish and releases toxins that kill the prey, which it can then feed on (*see p. 647*). No one is certain why these toxic blooms are becoming more frequent, but most phycologists believe that the blooms are caused by the continuous pumping of nutrients such as nitrogen and phosphorus into coastal waters. Sewage and agricultural runoff are probably the major sources. Another possibility is world trade: oceangoing ships are unintentionally trafficking in harmful algae, giving the algae a free ride to foreign ports and new habitats in which they can flourish. With people eating more seafood, this toxic menace in the world’s oceans will become increasingly more common in the future.

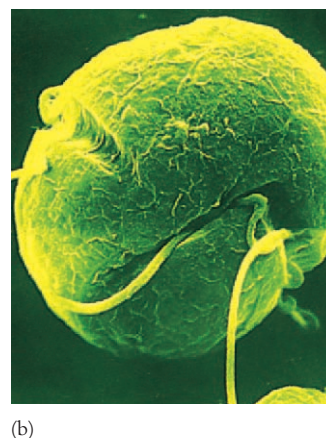
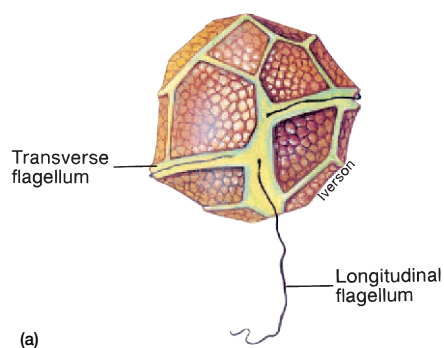


Figure 26.9 Dinoflagellates. (a) *Ceratium*. (b) Scanning electron micrograph of *Gymnodinium* ($\times 4,000$). Notice the plates of cellulose and the two flagella: one in the transverse groove and the other projecting outward.

Summary

1. Phycology (or algology) is the study of algae. Algae are eucaryotes that lack a well-developed vascular system and complex reproductive structures, but have chlorophyll and other pigments for carrying out oxygen-producing photosynthesis (or are closely related to photosynthetic species).
2. Algae are found in almost every environmental niche—in fresh, marine, and brackish water; in certain terrestrial environments; and on moist inanimate objects. They may be endosymbionts, parasites, or components of lichens, and make up a large part of the phytoplankton on the earth.
3. Classical algal classification is based primarily on cellular properties (**table 26.1**). Examples include cell wall structure and chemistry, nutrient storage forms, types of chlorophyll molecules and accessory pigments, flagella, morphology of the thallus, habitat, reproductive structures, and life history patterns (**table 26.2**). Molecular classification systems have shown the algae to be polyphyletic.
4. The vegetative structure of the algae varies from the relative simplicity of a single cell (**figure 26.1**) to the more striking complexity of organisms such as the giant kelps. Algae can be unicellular, colonial, filamentous, membranous, or tubular (**figure 26.2**).
5. Both sexual and asexual reproduction occur in the algae. Asexual reproduction includes fragmentation, the production of spores, and binary fission. In sexual reproduction gametes fuse to form a zygote.
6. Green algae (*Chlorophyta*) are a highly diverse group of organisms abundant in the sea, fresh water, and damp terrestrial habitats (**figure 26.3**).
7. Stoneworts/brittleworts (*Charophyta*) have more complex structures than the green algae. They contain whorls of short branches that arise regularly at their nodes. Their gametangia are complex and multicellular. Stoneworts are abundant in fresh to brackish water and are common as fossils.
8. Euglenoids (*Euglenophyta*) have chloroplasts that are biochemically similar to those of the green algae. They have a flexible proteinaceous pellicle inside the plasma membrane (**figure 26.5**).
9. The *Chrysophyta* contain golden-brown algae, yellow-green algae, and diatoms, and vary greatly in pigment composition, cell wall structure, and type of flagellated cells (**figure 26.6**).
10. Brown algae (*Phaeophyta*) are multicellular marine algae, some of which reach lengths of 75 m. The kelps—the largest of the brown algae—contribute greatly to the productivity of the sea as well as to many human needs (**figure 26.7**).
11. Red algae (*Rhodophyta*) have chlorophyll *a* and phycobilins and usually grow at great depths. Some produce agar (**figure 26.8**).
12. Dinoflagellates (*Pyrrophyta*) are unicellular motile algae that play a role in the red tides that are poisonous to many forms of life (**figure 26.9**).

Key Terms

algae 571
algology 571
amnesic shellfish poisoning 580
antheridium 574
aplanospore 573
benthic 571
binary fission 573
chrysolaminarin 577
ciguatera 580
diatom 577
diatomaceous earth 578
dinoflagellate 579
epitheca 577

euglenoid 576
fragmentation 573
frustule 577
hypotheca 577
kelp 578
laminarin 578
neustonic 571
oogonium 574
paralytic shellfish poisoning 580
pellicle 576
phycology 571
phytoplankton 571
plankton 571

planktonic 571
protothecosis 575
pyrenoid 573
red tides 580
scale 577
spore 573
stigma (eyespot) 575
stonewort 576
thallus 573
zooplankton 571
zoospore 573
zooxanthellae 579
zygote 574

Questions for Thought and Review

1. How can algae be distinguished from the photosynthetic bacteria?
2. What characteristics are used in algal classification?
3. Although multicellularity must have originated many times in the different multicellular groups, how do present-day green algae provide an especially clear example of the probable origin of multicellularity?
4. What characteristics do euglenoids share with higher plants? With animals? Why are they considered protists?
5. Why do marine algae vary so much more in shape and size than those found in fresh water?
6. The phytoplankton of the open oceans (e.g., the Sargasso Sea) are predominantly algae, and these cells form the base of oceanic food chains. Which morphological characteristics of algae are adaptations to a floating (planktonic) habitat?
7. Freshwater algae are distributed worldwide. They rapidly colonize artificial lakes and water impoundments. How do algae accomplish such widespread dispersal?
8. What problem do diatoms have with continued asexual reproduction? How is this problem solved?
9. How are the red algae similar to the cyanobacteria?
10. What are some important characteristics of the green algae? The red algae? The brown algae? The dinoflagellates?
11. Describe the distinguishing characteristics of each major algal group discussed in this chapter.

Critical Thinking Questions

1. Why do algae have so many different types of pigments for photosynthesis? Do these pigments serve other purposes?
2. Why don't we know as much about the basic biology of algae as we do about the biology of fungi, viruses, and bacteria?

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26.6 Algal Reproduction

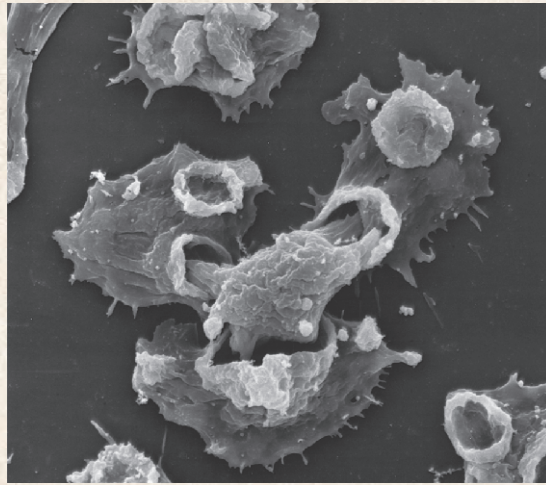
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CHAPTER 27

The Protozoa



This is a scanning electron micrograph ($\times 2,160$) of the protozoan *Naegleria fowleri*. Three *N. fowleri* from an axenic culture, attacking and beginning to devour or engulf a fourth, presumably dead amoeba, with their amoebastomes (suckerlike structures that function in phagocytosis). This amoeba is the major cause of the disease in humans called primary amebic meningoencephalitis.

Outline

- 27.1 Distribution 584
- 27.2 Importance 584
- 27.3 Morphology 585
- 27.4 Nutrition 586
- 27.5 Encystment and Excystment 586
- 27.6 Locomotory Organelles 586
- 27.7 Reproduction 586
- 27.8 Classification 587
- 27.9 Representative Types 588
 - Phylum *Sarcomastigophora* 588
 - Phylum *Labyrinthomorpha* 590
 - Phylum *Apicomplexa* 591
 - Phylum *Microspora* 591
 - Phylum *Ascetospora* 591
 - Phylum *Myxozoa* 591
 - Phylum *Ciliophora* 592

Concepts

1. Protozoa are protists exhibiting heterotrophic nutrition and various types of locomotion. They occupy a vast array of habitats and niches and have organelles similar to those found in other eucaryotic cells, and also specialized organelles.
2. Current protozoan taxonomy divides the protozoa into seven phyla: *Sarcomastigophora*, *Labyrinthomorpha*, *Apicomplexa*, *Microspora*, *Ascetospora*, *Myxozoa*, and *Ciliophora*. These phyla represent four major groups: flagellates, amoebae, ciliates, and sporozoa. In molecular classification schemes, the protozoa are polyphyletic eucaryotes.
3. Protozoa usually reproduce asexually by binary fission. Some have sexual cycles, involving meiosis and the fusion of gametes or gametic nuclei resulting in a diploid zygote. The zygote is often a thick-walled, resistant, and resting cell called a cyst. Some protozoa undergo conjugation in which nuclei are exchanged between cells.
4. All protozoa have one or more nuclei; some have a macro- and micronucleus.
5. Various protozoa feed by holophytic, holozoic, or saprozoic means; some are predatory or parasitic.

And a pleasant sight they are indeed. Their shapes range from teardrops to bells, barrels, cups, cornucopias, stars, snowflakes, and radiating suns, to the common amoebas, which have no real shape at all. Some live in baskets that look as if they were fashioned of exquisitely carved ivory filigree. Others use colored bits of silica to make themselves bright mosaic domes. Some even form graceful transparent containers shaped like vases or wine glasses of fine crystal in which they make their homes.

—Helena Curtis

Chapter 27 presents the major biological features of the protists known as protozoa. The most important groups are the flagellates, amoebae, sporozoa, and ciliates. The protists demonstrate the great adaptive potential of the basic single eucaryotic cell, as evidenced by their many nonrelated polyphyletic origins (**Phylogenetic Diagram 27**).

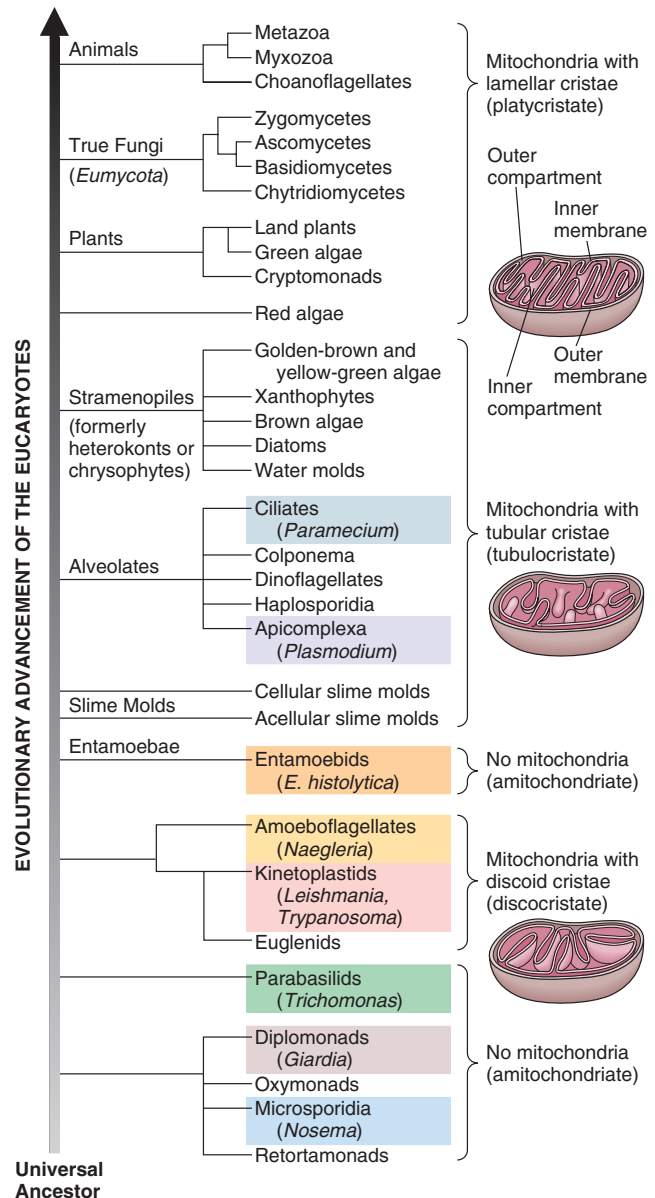
The microorganisms called **protozoa** [s., protozoan; Greek *protos*, first, and *zoon*, animal] are studied in the discipline called **protozoology**. A protozoan can be defined as a usually motile eucaryotic unicellular protist. Protozoa are directly related only on the basis of a single negative characteristic—they are not multicellular. All, however, demonstrate the basic body plan of a single protistan eucaryotic cell.

27.1 Distribution

Protozoa grow in a wide variety of moist habitats. Moisture is absolutely necessary for the existence of protozoa because they are susceptible to desiccation. Most protozoa are free living and inhabit freshwater or marine environments. Many terrestrial protozoa can be found in decaying organic matter, in soil, and even in beach sand; some are parasitic in plants or animals.

27.2 Importance

Protozoa play a significant role in the economy of nature. For example, they make up a large part of **plankton**—small, free-floating organisms that are an important link in the many aquatic food chains and food webs of aquatic environments. A **food chain** is a series of organisms, each feeding on the preceding one. A **food web** is a complex interlocking series of food chains. Protozoa are also useful in biochemical and molecular biological studies. Many biochemical pathways used by protozoa are present in all eucaryotic cells. Finally, some of the most important diseases of humans (see section 40.2) and animals (table 27.1) are caused by protozoa. [Microorganisms and ecosystems \(pp. 622–23\)](#)



Phylogenetic Diagram 27 **Tentative Phylogeny of the Protozoan-Like Eucaryotes Based on 18S rRNA Sequence Comparisons.** Recent molecular phylogeny of the nuclear SSU rRNA indicates that these eucaryotes are highly polyphyletic (protozoan groups are highlighted by different colors). Thus, like the algae, the protozoa do not represent a monophyletic group and the taxon “Protozoa” should not be used in classification schemes that seek to represent true molecular evolutionary histories. The word protozoa can still be used (as it is in this chapter) to denote a nonrelated polyphyletic group of eucaryotic organisms that share some morphological, reproductive, ecological, and biochemical characteristics.

Table 27.1 Pathogenic Protozoa That Cause Major Diseases of Domestic Animals

Protozoan Group ^a	Genus	Host	Preferred Site of Infection	Disease
Amoebae	<i>Entamoeba</i>	Mammals	Intestine	Amebiasis
	<i>Iodamoeba</i>	Swine	Intestine	Enteritis
Sporozoa	<i>Babesia</i>	Cattle	Blood cells	Babesiosis
	<i>Theileria</i>	Cattle, sheep, goats	Blood cells	Theileriasis
	<i>Sarcocystis</i>	Mammals, birds	Muscles	Sarcosporidiosis
	<i>Toxoplasma</i>	Cats	Intestine	Toxoplasmosis
	<i>Isospora</i>	Dogs	Intestine	Coccidiosis
	<i>Eimeria</i>	Cattle, cats, chickens, swine	Intestine	Coccidiosis
	<i>Plasmodium</i>	Many animals	Bloodstream, liver	Malaria
	<i>Leucocytozoon</i>	Birds	Spleen, lungs, blood	Leucocytozoonosis
	<i>Cryptosporidium</i>	Mammals	Intestine	Cryptosporidiosis
	<i>Balantidium</i>	Swine	Large intestine	Balantidiasis
Ciliates	<i>Leishmania</i>	Dogs, cats, horses, sheep, cattle	Spleen, bone marrow, mucous membranes	Leishmaniasis
Flagellates	<i>Trypanosoma</i>	Most animals	Blood	Trypanosomiasis
	<i>Trichomonas</i>	Horses, cattle	Genital tract	Trichomoniasis (abortion)
	<i>Histomonas</i>	Birds	Intestine	Blackhead disease
	<i>Giardia</i>	Mammals	Intestine	Giardiasis

^aThese groups are distinguished from one another largely by their mechanism of locomotion (see text).

27.3 Morphology

Because protozoa are eucaryotic cells, in many respects their morphology and physiology are the same as the cells of multicellular animals (see figures 4.2 and 4.3). However, because all of life's various functions must be performed within the individual protozoan, some morphological and physiological features are unique to protozoan cells. In some species the cytoplasm immediately under the plasma membrane is semisolid or gelatinous, giving some rigidity to the cell body. It is termed the **ectoplasm**. The bases of the flagella or cilia and their associated fibrillar structures are embedded in the ectoplasm. The plasma membrane and structures immediately beneath it are called the **pellicle**. Inside the ectoplasm is the area referred to as the **endoplasm**, which is more fluid and granular in composition and contains most of the organelles. Some protozoa have one nucleus, others have two or more identical nuclei. Still other protozoa have two distinct types of nuclei—a macronucleus and one or more micronuclei. The **macronucleus**, when present, is typically larger and associated with trophic activities and regeneration processes. The **micronucleus** is diploid and involved in both genetic recombination during reproduction and the regeneration of the macronucleus.

One or more vacuoles are usually present in the cytoplasm of protozoa. These are differentiated into contractile, secretory, and food vacuoles. **Contractile vacuoles** function as osmoregulatory organelles in those protozoa that live in a hypotonic environment,

such as a freshwater lake. Osmotic balance is maintained by continuous water expulsion. Most marine protozoa and parasitic species are isotonic to their environment and lack such vacuoles. **Phagocytic vacuoles** are conspicuous in holozoic and parasitic species and are the sites of food digestion (see figure 4.10). **Secretory vacuoles** usually contain specific enzymes that perform various functions (such as excystation).

Most anaerobic protozoa (such as *Trichonympha*, which lives in the gut of termites; see figure 28.26) have no mitochondria, no cytochromes, and an incomplete tricarboxylic acid cycle. However, some do have small, membrane-delimited organelles termed **hydrogenosomes**. These structures contain a unique electron transfer pathway in which hydrogenase transfers electrons to protons (which act as the terminal electron acceptors), and molecular hydrogen is formed. Other protozoa have mitochondria with discoid cristae (trypanosomes), tubular mitochondrial cristae (ciliates, sporozoa), and lamellar cristae (foraminiferans).

1. Describe a typical protozoan.
2. What roles do protozoa play in the trophic structure of their communities and in the organisms with which they associate?
3. What is unique about the nuclei of some protozoa?
4. Where can protozoa be found?
5. What are the functions of contractile, phagocytic, and secretory vacuoles?

27.4 Nutrition

Most protozoa are chemoheterotrophic. Two types of heterotrophic nutrition are found in the protozoa: holozoic and saprozoic. In **holozoic nutrition**, solid nutrients such as bacteria are acquired by phagocytosis and the subsequent formation of phagocytic vacuoles. Some ciliates have a specialized structure for phagocytosis called the **cytostome** (cell mouth). In **saprozoic nutrition**, soluble nutrients such as amino acids and sugars cross the plasma membrane by pinocytosis, diffusion, or carrier-mediated transport (facilitated diffusion or active transport).

27.5 Encystment and Excystment

Many protozoa are capable of **encystation**. They develop into a resting stage called a **cyst**, which is a dormant form marked by the presence of a wall and by very low metabolic activity. Cyst formation is particularly common among aquatic, free-living protozoa and parasitic forms. Cysts serve three major functions: (1) they protect against adverse changes in the environment, such as nutrient deficiency, desiccation, adverse pH, and low partial pressure of O₂; (2) they are sites for nuclear reorganization and cell division (reproductive cysts); and (3) they serve as a means of transfer between hosts in parasitic species.

Although the exact stimulus for **excystation** (escape from the cysts) is unknown, excystation generally is triggered by a return to favorable environmental conditions. For example, cysts of parasitic species excyst after ingestion by the host and form the vegetative form called the **trophozoite**.

27.6 Locomotory Organelles

A few protozoa are nonmotile. Most, however, can move by one of three major types of locomotory organelles: pseudopodia, flagella, or cilia. **Pseudopodia** [s., pseudopodium; false feet] are cytoplasmic extensions found in the amoebae that are responsible for the movement and food capture. There are many types of pseudopodia. Flagellates and ciliates move by flagella and cilia. Electron microscopy has shown that protozoan flagella and cilia are structurally the same and identical in function to those of other eucaryotic cells (see figures 4.22–4.25).

27.7 Reproduction

Most protozoa reproduce asexually, and some also carry out sexual reproduction. The most common method of asexual reproduction is **binary fission**. During this process the nucleus first undergoes mitosis, then the cytoplasm divides by cytokinesis to form two identical individuals (**figure 27.1**).



Figure 27.1 Protozoan Reproduction. Binary fission in *Paramecium caudatum* ($\times 100$).

The most common method of sexual reproduction is **conjugation**. In this process there is an exchange of gametes between paired protozoa of complementary mating types (**conjugants**; see figure 2.13b). Conjugation is most prevalent among ciliate protozoa. A well-studied example is *Paramecium caudatum* (**figure 27.2**). At the beginning of conjugation, two ciliates unite, fusing their pellicles at the contact point. The macronucleus in each is degraded. The individual micronuclei divide twice by meiosis to form four haploid pronuclei, three of which disintegrate. The remaining pronucleus divides again mitotically to form two gametic nuclei, a stationary one and a migratory one. The migratory nuclei pass into the respective conjugates. Then the ciliates separate, the gametic nuclei fuse, and the resulting diploid zygote nucleus undergoes three rounds of mitosis. The eight resulting nuclei have different fates: one nucleus is retained as a micronucleus; three others are destroyed; and the four remaining nuclei develop into macronuclei. Each separated conjugant now undergoes cell division. Eventually progeny with one macronucleus and one micronucleus are formed.

1. What specific nutritional types exist among protozoa?
2. What functions do cysts serve for a typical protozoan? What causes excystation to occur?
3. What is a pseudopodium?
4. How do protozoa reproduce asexually?
5. How do protozoa reproduce sexually? Describe the process of ciliate conjugation.

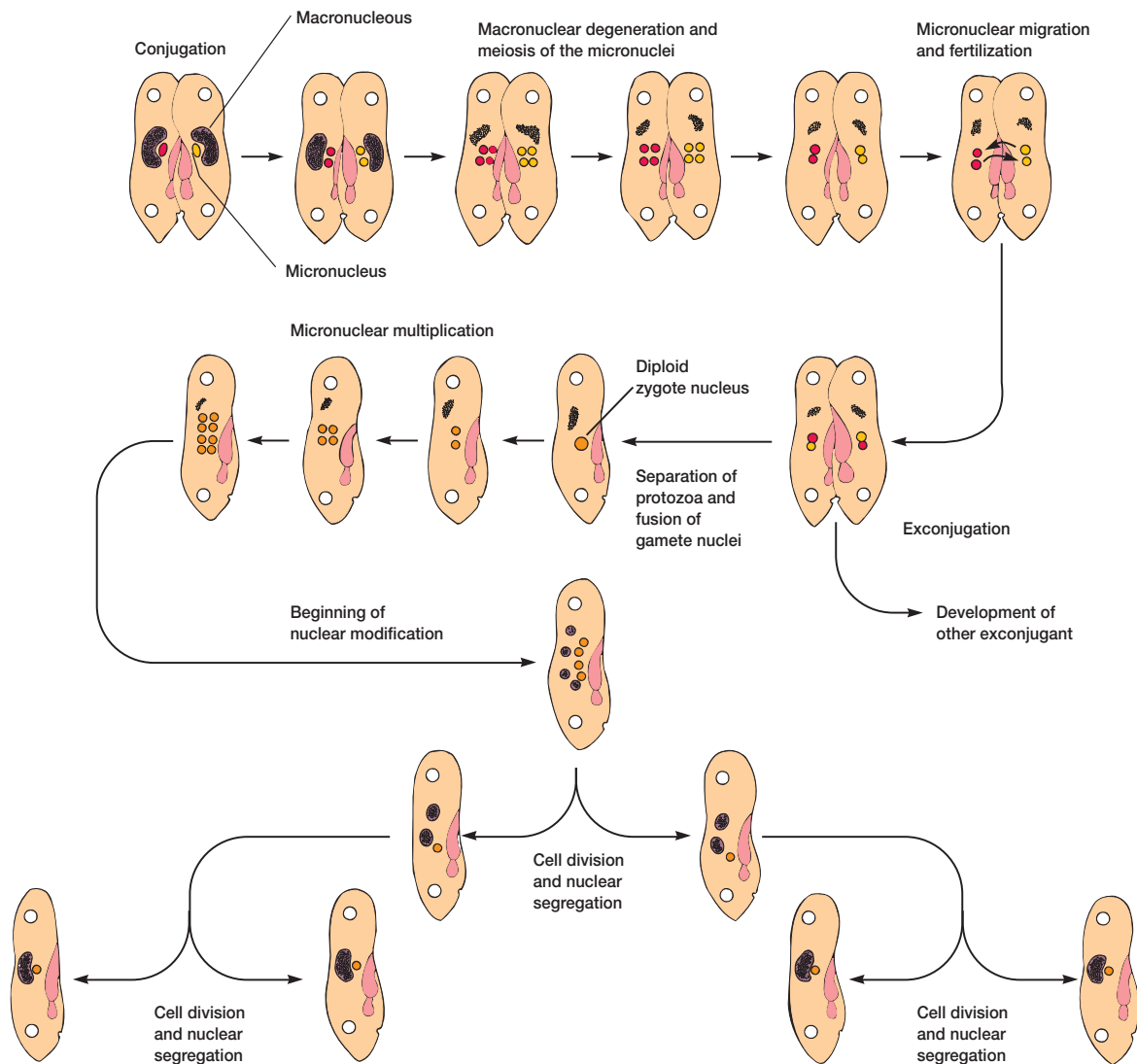


Figure 27.2 Conjugation in *Paramecium caudatum*, Schematic Drawing. Follow the arrows. After the conjugants separate, only one of the exconjugants is followed; however, a total of eight new protozoa result from the conjugation.

27.8 Classification

Many protozoan taxonomists regard the *Protozoa* as a subkingdom, which contains seven of the 14 phyla found within the kingdom *Protista* (table 27.2). The phylum *Sarcomastigophora* consists of flagellates and amoebae with a single type of nucleus. The phyla *Labyrinthomorpha*, *Apicomplexa*, *Microspora*, *Asce-*

topora, and *Myxozoa* have either saprozoic or parasitic species. The phylum *Ciliophora* has ciliated protozoa with two types of nuclei. The classification of this subkingdom into phyla is based primarily on types of nuclei, mode of reproduction, and mechanism of locomotion.

More recent classifications are quite different. In 1993 T. Cavalier-Smith proposed that the protozoa be elevated to

Table 27.2 Abbreviated Classification of the Subkingdom *Protozoa*^a

Taxonomic Group	Characteristics	Examples
Phylum: <i>Sarcomastigophora</i>	Locomotion by flagella, pseudopodia, or both; when present, sexual reproduction is essentially syngamy (union of gametes external to the parents); single type of nucleus	
Subphylum: <i>Mastigophora</i>	One or more flagella; division by longitudinal binary fission; sexual reproduction in some groups	
Class: <i>Zoomastigophorea</i>	Chromatophores absent; one to many flagella; amoeboid forms, with or without flagella; sexuality known in some groups; mainly parasitic	<i>Trypanosoma</i> <i>Giardia</i> <i>Trichomonas</i> <i>Leishmania</i> <i>Trichonympha</i>
Subphylum: <i>Sarcodina</i>	Locomotion primarily by pseudopodia; shells (tests) often present; flagella restricted to reproductive stages when present; asexual reproduction by fission; mostly free living	
Superclass: <i>Rhizopoda</i>	Locomotion by pseudopodia or by protoplasmic flow with discrete pseudopodia; some contain tests	<i>Amoeba</i> <i>Elphidium</i> <i>Coccolodiscus</i> <i>Labyrinthula</i>
Phylum: <i>Labyrinthomorpha</i>	Spindle-shaped cells capable of producing mucous tracks; trophic stage as ectoplasmic network; nonamoeboid cells; saprozoic and parasitic on algae and seagrass	
Phylum: <i>Apicomplexa</i>	All members have a spore-forming stage in their life cycle; contain an apical complex; sexuality by syngamy; all species parasitic; cysts often present; cilia absent; often called the Sporozoa	<i>Plasmodium</i> <i>Toxoplasma</i> <i>Eimeria</i> <i>Cryptosporidium</i> <i>Nosema</i> <i>Haplosporidium</i>
Phylum: <i>Microspora</i>	Unicellular spores with spiroplasm containing polar filaments; obligatory intracellular parasites	
Phylum: <i>Asctosporia</i>	Spore with one or more spiroplasm; no polar capsules or polar filaments; all parasitic in invertebrates	
Phylum: <i>Myxozoa</i>	Spores of multicellular origin; one or more polar capsules; all parasitic, especially in fish	<i>Myxosoma</i>
Phylum: <i>Ciliophora</i>	Simple cilia or compound ciliary organelles in at least one stage in the life cycle; two types of nuclei; contractile vacuole present; binary fission transverse; sexuality involving conjugation; most species free living, but many commensal, some parasitic	<i>Didinium</i> <i>Stentor</i> <i>Vorticella</i> <i>Tetrahymena</i> <i>Paramecium</i> <i>Tokophrya</i> <i>Entodinium</i> <i>Nyctotherus</i> <i>Balantidium</i> <i>Ichthyophthirius</i>

^aBased on the 1980 Committee on Systematics and Evolution of the Society of Protozoologists.

kingdom status with 18 phyla based on the structure of mitochondrial cristae and other characteristics (see section 19.7). The acceptance of this new classification by protozoologists, however, remains to be determined. In recent molecular classification schemes, the protozoa do not exist as a discrete taxon. Protozoan-like eucaryotes are found at all evolutionary levels (Phylogenetic Diagram 27).

27.9 Representative Types

This section describes some representatives of each group of protozoan protists to present an overview of protozoan diversity and to provide a basis for comparing different groups. For simplicity, the standard classification summarized in table 27.2 will be followed.

Phylum *Sarcomastigophora*

Protists that have a single type of nucleus and possess flagella (subphylum *Mastigophora*) or pseudopodia (subphylum *Sarcodina*) are placed in the phylum *Sarcomastigophora*. Both sexual and asexual reproduction are seen in this phylum.

The subphylum *Mastigophora* contains both phytoflagellates, chloroplast-bearing flagellates and close relatives, and **zooflagellates**. Zooflagellates do not have chlorophyll and are either holozoic, saprozoic, or symbiotic. Asexual reproduction occurs by longitudinal binary fission along the major body axis. Sexual reproduction is known for a few species, and encystment is common. Zooflagellates are characterized by the presence of one or more flagella. Most members are uninucleate. One major group, the kinetoplastids, has its mitochondrial DNA in a special region called the **kinetoplast** (figure 27.3a; see also figures 2.13c and 4.13).

Some zooflagellates are free living. The choanoflagellates are a distinctive example in that they have one flagellum, are solitary or colonial, and are on stalks. Other zooflagellates form symbiotic relationships. For example, *Trichonympha* species (see figure 28.26) are found in the intestine of termites and produce enzymes that the termite needs to digest the wood particles on which it feeds. The protozoan-termite relationship (p. 598)

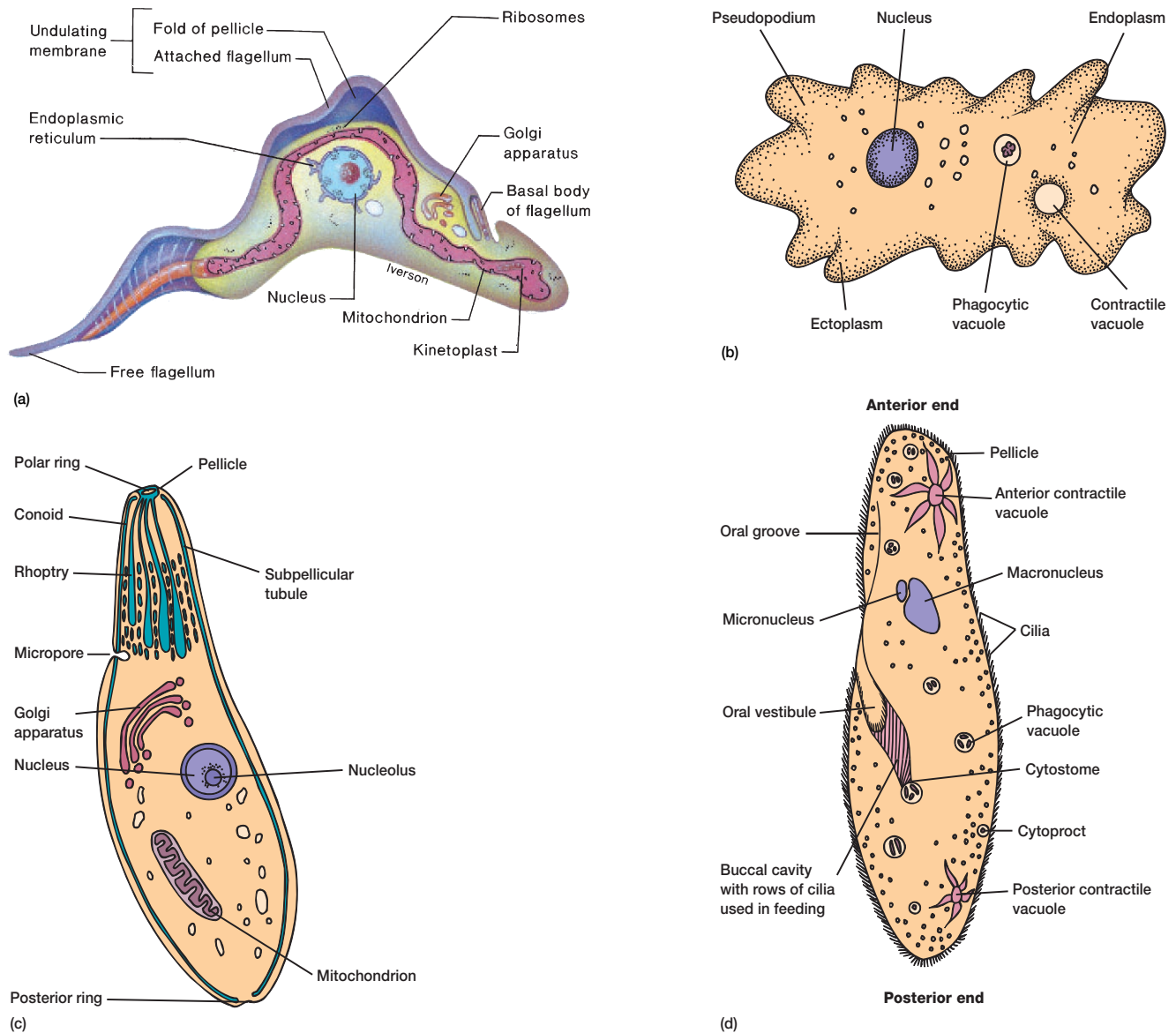


Figure 27.3 Drawings of Some Representative Protozoa. (a) Structure of the flagellate, *Trypanosoma brucei rhodesiense*. (b) The structure of the amoeboid protist, *Amoeba proteus*. (c) Structure of an apicomplexan sporozoite. (d) Structure of the ciliate *Paramecium caudatum*.

Many zooflagellates are important human parasites. *Giardia lamblia* (see figure 40.18) can be found in the human intestine where it may cause severe diarrhea. It is transmitted through water that has been contaminated with feces (see section 40.2). Trichomonads, such as *Trichomonas vaginalis*, live in the vagina and urethra of women and in the prostate, seminal vesicles, and urethra of men. They are transmitted primarily by sexual intercourse (See table 39.4 on page 927 for a summary of all the sexually transmitted diseases covered in this textbook.) [Giardiasis \(pp. 953–54\)](#). [Trichomoniasis \(p. 958\)](#)

The zooflagellates called **trypanosomes** are important blood pathogens of humans and animals in certain parts of the world. Because they live in the blood, they are also called hemoflagellates. These parasites (figure 27.3a) have a typical zooflagellate structure and appear to be the earliest diverging branch of protists with mitochondria and peroxisomes. A major human trypanosomal disease is African sleeping sickness caused by *Trypanosoma brucei rhodesiense* or *T. brucei gambiense* (see section 40.2).

The subphylum *Sarcodina* contains the amoeboid protists. They are found throughout the world in both fresh and salt water and

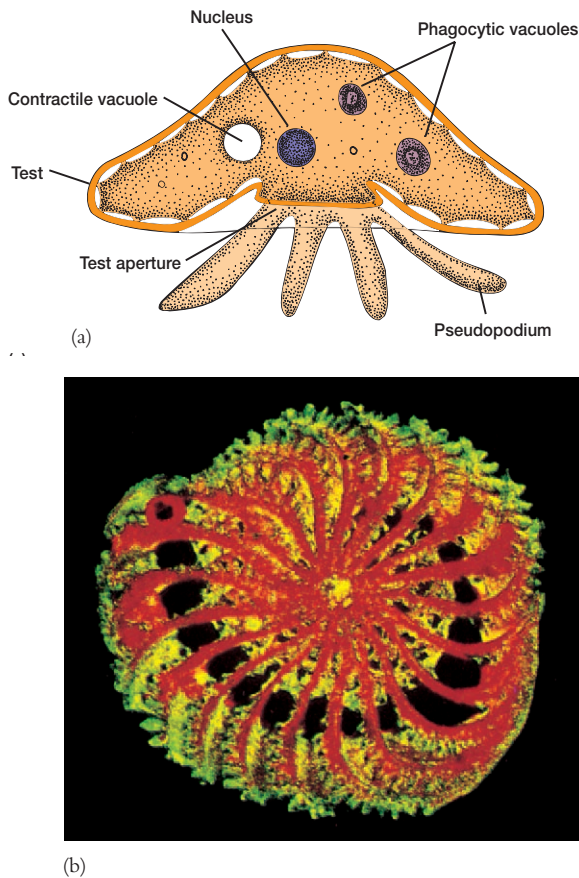


Figure 27.4 Some Free-living Sarcodines. (a) An illustration of *Arcella*, showing the test or shell that is made of chitinlike material secreted by the protist. (b) The test of the foraminiferan, *Elphidium cristum* ($\times 100$). (c) A group of siliceous radiolarian shells, light micrograph ($\times 63$).

are abundant in the soil. Several species are parasites of mammals. Simple amoebae [s., amoeba] move almost continually using their pseudopodia (**amoeboid movement**). Many have no definite shape, and their internal structures (figure 27.3b) occupy no particular position. The single nucleus, contractile and phagocytic vacuoles, and ecto- and endoplasm shift as the amoebae move. Amoebae engulf a variety of materials (small algae, bacteria, other protozoa) through phagocytosis. Some material moves into and out of the plasma membrane by pinocytosis. Reproduction in the amoebae is by simple asexual binary fission. Some amoebae can form cysts.

Many free-living forms are more complex than simple amoebae. *Arcella* manufactures a loose-fitting shell or **test** for protection (figure 27.4a). These amoebae extend their pseudopodia from the test aperture to either feed or creep along. The foraminiferans and radiolarians primarily are marine amoebae, with a few occurring in fresh and brackish water. Most foraminiferans live on the seafloor, whereas radiolarians are usually found in the open sea (**Box 27.1**). Foraminiferan tests and radiolarian skeletons have many unique and beautiful shapes (figure 27.4b,c). They range in diameter from about 20 μm to several cm.

Finally, there are many symbiotic amoebae, most of which live in other animals. Two common genera are *Endamoeba* and

Entamoeba. *Endamoeba blattae* is common in the intestine of cockroaches, and related species are present in termites. *Entamoeba histolytica* (see figure 40.17) is an important parasite of humans, in whom it often produces severe amoebic dysentery, which may be fatal. Free-living amoebae from two genera, *Naegleria* and *Acanthamoeba*, can cause disease in humans and other mammals (chapter opening figure; see also section 40.2).

1. What characteristics would be exhibited by a protozoan that belongs to the phylum *Sarcomastigophora*? What two subphyla does it contain?
2. How would you characterize a zooflagellate? An amoeba?
3. What two human diseases are caused by zooflagellates?
4. Where can two different symbiotic amoebae be found?

Phylum Labyrinthomorpha

The very small phylum *Labyrinthomorpha* consists of protists that have spindle-shaped or spherical nonamoeboid vegetative cells. In some genera, amoeboid cells move within a network of mucous

Box 27.1

The Importance of Foraminiferans

Of over 40,000 described species of foraminiferans, about 90% are fossil. During the Tertiary period (about 230 million years ago), the foraminiferans contributed massive shell accumulations to geologic formations. They were so abundant that they formed thick deposits, which became uplifted over time and exposed as great beds of limestone in Europe, Asia, and Africa. The White Cliffs of Dover, the famous landmark of southern England, are made up almost entirely

of foraminiferan shells. The Egyptian pyramids of Gizeh, near Cairo, are built of foraminiferan limestone. Currently, foraminiferans are important aids to geologists in identifying and correlating rock layers as they search for oil-bearing strata. The calcareous shells of abundant planktonic foraminiferans are today settling and accumulating over much of the ocean floor as thick deposits called “globigerina ooze”—limestone of the distant future.

tracks using a typical gliding motion. Most members are marine and either saprozoic or parasitic on algae. Several years ago *Labyrinthula* killed most of the “eel grass” on the Atlantic coast, depriving ducks of their food and starving many of them.

Phylum Apicomplexa

The **apicomplexans**, often collectively called the sporozoans, have a spore-forming stage in their life cycle and lack special locomotory organelles (except in the male gametes, and the zygote or ookinete). They are either intra- or intercellular parasites of animals and are distinguished by a unique arrangement of fibrils, microtubules, vacuoles, and other organelles, collectively called the apical complex, which is located at one end of the cell.

The **apical complex** contains several components (figure 27.3c). One or two polar rings are at the apical end. The **conoid** consists of a cone of spirally arranged fibers lying next to the polar rings. Subpellicular microtubules radiate from the polar rings and probably serve as support elements. Two or more **rhoptries** extend to the plasma membrane and secrete their contents at the cell surface. These secretions aid in the penetration of the host cell. One or more micropores are thought to function in the intake of nutrients.

Apicomplexans have complex life cycles in which certain stages occur in one host (the mammal) and other stages in a different host (often a mosquito). The life cycle has both asexual and sexual phases and is characterized by an alternation of haploid and diploid generations. At some point an asexual reproduction process called schizogony occurs. **Schizogony** is a rapid series of mitotic events producing many small infective organisms through the formation of uninuclear buds. Sexual reproduction involves the fertilization of a large female macrogamete by a small, flagellated male gamete. The resulting zygote becomes a thick-walled cyst called an **oocyst**. Within the oocyst, meiotic divisions produce infective haploid spores.

The four most important sporozoan parasites are *Plasmodium* (the causative agent of malaria), *Cryptosporidium* (the causative agent of cryptosporidiosis), *Toxoplasma* (the causative agent of toxoplasmosis), and *Eimeria* (the causative agent of coccidiosis). [Malaria \(pp. 954–56\)](#)

Phylum Microspora

The small microsporans (3 to 6 μm) are obligatory intracellular parasites lacking mitochondria. The infective stage is transmitted from host to host as a resistant spore. Included in these protozoa are several species of some economic importance because they parasitize beneficial insects. *Nosema bombycis* parasitizes silkworms (figure 27.5) causing the disease **pebrine**, and *Nosema apis* causes serious dysentery (foul brood) in honeybees. There has been an increased interest in these parasites because of their possible role as biological control agents for certain insects. For example, *Nosema locustae* has been approved and registered by the United States Environmental Protection Agency for use in long-lasting control of rangeland grasshoppers. Recently seven microsporidian genera (*Nosema*, *Encephalitozoon*, *Pleistophora*, *Microsporidium*, *Vittaforma*, *Trachipleistophora*, and *Enterocytozoon*) have been implicated in human diseases in immunosuppressed and AIDS patients.

Phylum Ascetospora

Ascetospora is a relatively small phylum that consists exclusively of parasitic protists characterized by spores lacking polar caps or polar filaments. Ascetosporans such as *Haplosporidium* are parasitic primarily in the cells, tissues, and body cavities of mollusks.

Phylum Myxozoa

The myxozoans are all parasitic, most on freshwater and marine fish. They have a resistant spore with one to six coiled polar filaments. The most economically important myxozoan is *Myxosoma cerebralis*, which infects the nervous system and auditory organ of trout and salmon (salmonids). Infected fish lose their sense of balance and tumble erratically—thus the name whirling or tumbling disease. **Proliferative kidney disease**, caused by an unclassified myxozoan, has become one of the most important diseases of cultured salmon throughout the world.

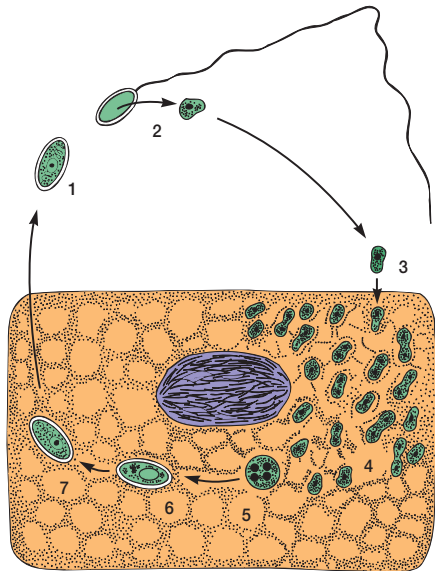


Figure 27.5 The Microsporean *Nosema bombycis*, Which is Fatal to Silkworms. (1) A typical spore with one coiled filament. (2) When ingested, it extrudes the filament. (3) The parasite enters an epithelial cell in the intestine of the silkworm and (4) divides many times to form small amoebae that eventually fill the cell and kill it. During this phase, some of the amoebae with four nuclei become spores (5, 6, 7). Silkworms are infected by eating leaves contaminated by the feces of infected worms.

1. Describe a typical apicomplexan (sporozoan) protist, including its apical complex.
2. Summarize the sporozoan life cycle. What is schizogony?
3. What are the four most important sporozoan parasites and the diseases they cause?
4. Give one economically important disease caused by a microsporidian. What group of animals do myxosporidians usually parasitize?

Phylum Ciliophora

The phylum *Ciliophora* is the largest of the seven protozoan phyla. There are about 8,000 species of these unicellular, heterotrophic protists that range from about 10 to 3,000 μm long. As their name implies, ciliates employ many cilia as locomotory organelles. The cilia are generally arranged either in longitudinal rows (figure 27.3d; see also figure 4.24) or in spirals around the body of the organism. They beat with an oblique stroke; therefore the protist revolves as it swims. Coordination

of ciliary beating is so precise that the protist can go either forward or backward.

There is great variation in ciliate shape, and most do not look like the slipper-shaped *Paramecium* (see figures 2.8e and 4.1a). In some species (*Vorticella*) the protozoan attaches itself to the substrate by a long stalk. *Stentor* attaches to a substrate and stretches out in a trumpet shape to feed (see figure 4.1e). A few species have tentacles for the capture of prey. Some can discharge toxic thread-like darts called toxicysts, which are used in capturing prey.

A most striking feature of ciliates is their ability to capture many particles in a short time by the action of the cilia around the buccal cavity. Food first enters the cytostome and passes into phagocytic vacuoles that fuse with lysosomes after detachment from the cytostome. A vacuole's contents are digested when the vacuole is acidified and lysosomes release digestive enzymes into it. After the digested material has been absorbed into the cytoplasm, the vacuole fuses with a special region of the pellicle called the **cytoproct** and empties its waste material to the outside. Contractile vacuoles are used for osmoregulation and are present chiefly in freshwater species.

Most ciliates have two types of nuclei: a large macronucleus and a smaller micronucleus. The micronucleus is diploid and contains the normal somatic chromosomes. It divides by mitosis and transmits genetic information through meiosis and sexual reproduction. Macronuclei are derived from micronuclei by a complex series of steps. Within the macronucleus are many chromatin bodies, each containing many copies of only one or two genes. Macronuclei are thus polyploid and divide by elongating and then by constricting. They produce mRNA to direct protein synthesis, maintain routine cellular functions, and control normal cell metabolism.

Some ciliates reproduce asexually by transverse binary fission, forming two equal daughter protozoa. Many ciliates also reproduce by conjugation as previously described.

Although most ciliates are free living, symbiotic forms do exist. Some ciliated protozoa live as harmless commensals. For example, *Entodinium* is found in the rumen of cattle, and *Nyctotherus* occurs in the colon of frogs. Other ciliates are strict parasites. For example, *Balantidium coli* lives in the intestine of mammals, including humans, where it can produce dysentery. *Ichthyophthirius* lives in freshwater where it can attack many species of fish, producing a disease known as "ick."

1. Describe the morphology of a typical ciliated protozoan.
2. Describe the food-gathering structures found in the ciliated protozoa.
3. In ciliates, what is the function of the macronucleus? The micronucleus?
4. How do ciliates reproduce?
5. Where can the following ciliates be found: *Entodinium*, *Nyctotherus*, *Balantidium*, *Ichthyophthirius*?

Summary

1. Protozoa are protists that can be defined as usually motile eucaryotic unicellular microorganisms.
2. Protozoa are found wherever other eucaryotic organisms exist. They are important components of food chains and food webs. Many are parasitic in humans and animals (**table 27.1**), and some have become very useful in the study of molecular biology.
3. Because protozoa are eucaryotic cells, in many respects their morphology and physiology resemble those of multicellular animals. However, because all their functions must be performed within the individual protist, many morphological and physiological features are unique to protozoan cells.
4. Some protozoa can secrete a resistant covering and go into a resting stage (encystation) called a cyst. Cysts protect the organism against adverse environments, function as a site for nuclear reorganization, and serve as a means of transmission in parasitic species.
5. Protozoa move by one of three major types of locomotory organelles: pseudopodia, flagella, or cilia. Some have no means of locomotion.
6. Most protozoa reproduce asexually (**figure 27.1**), some use sexual reproduction (**figure 27.2**), and some employ both methods.
7. According to the classical classification scheme, there are seven protozoan phyla (**table 27.2**). The phylum *Sarcomastigophora* is characterized by protists that have a single type of nucleus, possess flagella (subphylum *Mastigophora*), pseudopodia (subphylum *Sarcodina*), or both types of locomotory organelles.
8. The subphylum *Sarcodina* consists of the amoeboid protists. They are found throughout the world in both fresh and salt water and in the soil. Some species are parasitic.
9. The phylum *Labyrinthomorpha* contains protists that have spindle-shaped or spherical nonamoeboid vegetative cells. Most members are marine and either saprozoic or parasitic on algae.
10. The phylum *Apicomplexa* consists of sporozoan protists that possess an apical complex, a unique arrangement of fibrils, microtubules, vacuoles, and other organelles at one end of the cell. Representative members include the *Plasmodium* parasites, which cause malaria; *Toxoplasma*, which causes toxoplasmosis; and *Eimeria*, the agent of coccidiosis.
11. The phylum *Microspora* consists of very small protists that are intracellular parasites of every major animal group. They are transmitted from one host to the next as a spore, the form from which the group obtains its name (**figure 27.5**).
12. The phylum *Ascetospora* contains protists that produce spores lacking polar capsules. These protists are primarily parasitic in mollusks.
13. The phylum *Myxozoa* consists entirely of parasitic species, usually found in fish. The spore is characterized by one to six polar filaments.
14. The phylum *Ciliophora* comprises a group of protists that have cilia and two types of nuclei. Conjugation in the *Ciliophora* is a form of sexual reproduction that involves exchange of micronuclear material.

Key Terms

amoeboid movement 590
 apical complex 591
 apicomplexan 591
 binary fission 586
 conjugant 586
 conjugation 586
 conoid 591
 contractile vacuole 585
 cyst 586
 cytoproct 592
 cytostome 586
 ectoplasm 585
 encystation 586

endoplasm 585
 excystation 586
 food chain 584
 food web 584
 holozoic nutrition 586
 hydrogenosome 585
 kinetoplast 588
 macronucleus 585
 micronucleus 585
 oocyst 591
 pebrine 591
 pellicle 585
 phagocytic vacuole 585

plankton 584
 proliferative kidney disease 591
 protozoa 584
 protozoology 584
 pseudopodia 586
 rhoptry 591
 saprozoic nutrition 586
 schizogony 591
 secretory vacuole 585
 test 590
 trophozoite 586
 trypanosome 589
 zooflagellate 588

Questions for Thought and Review

1. What is the economic impact or human relevance of the protozoa?
2. What criteria are now used in the classification of protozoa?
3. Seven protozoan phyla have been discussed. What are their distinguishing characteristics?
4. What are some typical organelles found in the protozoa?
5. What advantage does cyst formation give to protozoa?
6. How do protozoa move? Reproduce?
7. If the diversity within the seven protozoan phyla is considered, which one shows the greatest evolutionary advancement? Defend your answer.
8. The protozoa are said to be grouped together for a negative reason. What does this mean?
9. Describe how DNA is distributed to daughter cells when the ciliate *Paramecium* divides. Include a discussion of both conjugation and binary fission.
10. How does a protozoan cyst differ from a bacterial endospore?

Critical Thinking Questions

1. Why don't we know as much about the basic biology of protozoa as we know about the biology of fungi, viruses, and bacteria?
2. The text suggests that excystation requires the recognition of environmental signals. Following this line of thinking, suggest some antiprotozoan targets that could be exploited to prevent excystation in patients who have ingested cysts. Once the cysts emerge, are there other targets?
3. Suggest a reason or mechanism why, in some protozoa, the cytoplasmic material (ectoplasm) just under the plasma membrane is so rigid.

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PART VIII

Ecology and Symbiosis

Chapter 28

Microorganism Interactions and
Microbial Ecology

Chapter 29

Microorganisms in Aquatic
Environments

Chapter 30

Microorganisms in Terrestrial
Environments

CHAPTER 28

Microorganism Interactions and Microbial Ecology



Microorganisms living in environments where most known organisms cannot survive are important for understanding microbial diversity. These strands of iron-oxidizing *Ferroplasma* were discovered growing at pH 0 in an abandoned mine near Redding, California. This hardy microorganism has only a plasma membrane to protect itself from the rigors of this harsh environment.

Outline

- | | | | |
|------|---------------------------------------|------|---|
| 28.1 | Foundations of Microbial Ecology 596 | 28.4 | The Physical Environment 619 |
| 28.2 | Microbial Interactions 596 | | The Microenvironment and Niche 619 |
| | Mutualism 598 | | Biofilms and Microbial Mats 620 |
| | Protocooperation 604 | | Microorganisms and Ecosystems 622 |
| | Commensalism 606 | | Microorganism Movement between Ecosystems 623 |
| | Predation 607 | | Stress and Ecosystems 624 |
| | Parasitism 609 | 28.5 | Methods in Microbial Ecology 626 |
| | Amensalism 609 | | |
| | Competition 609 | | |
| | Symbioses in Complex Systems 610 | | |
| 28.3 | Nutrient Cycling Interactions 611 | | |
| | Carbon Cycle 611 | | |
| | Sulfur Cycle 614 | | |
| | Nitrogen Cycle 615 | | |
| | Iron Cycle 616 | | |
| | Manganese Cycle 617 | | |
| | Other Cycles and Cycle Links 617 | | |
| | Microorganisms and Metal Toxicity 618 | | |

Concepts

1. Most microorganisms in complex communities have not been grown or characterized. This has limited our understanding of microorganism interactions and their roles in nature and disease. Molecular techniques are providing a better understanding of these uncultured organisms.
2. The term symbiosis, or “together-life,” can be used to describe many of the interactions between microorganisms, and also microbial interactions with higher organisms, including plants and animals. These interactions may be positive or negative.
3. Microbial ecology is the study of microbial relationships with other organisms and also with their nonliving environments. These relationships, based on interactive uses of resources, have effects extending to the global scale.
4. Positive interactions include mutualism, proto cooperation, and commensalism. Negative interactions include parasitism, predation, amensalism, and competition. These interactions are important in natural processes and in the occurrence of disease. The interactions can vary depending on the environment and changes in the interacting organisms.
5. Microorganisms, as they interact, can form complex physical assemblages that are often described as biofilms. These occur on living and inert surfaces and have major impacts on microbial survival and the occurrence of disease.
6. Microorganisms also interact by the use of chemical signal molecules, which allow the microbial population to respond to increased population density. Such responses include quorum sensing, which controls a wide variety of microorganism properties.
7. Energy, electrons, and nutrients must be available in a suitable physical environment for microorganisms to function. Microbes interact with their environment to obtain energy (from light or chemical sources), electrons, and nutrients, which leads to a process called biogeochemical cycling. Microorganisms change the physical state and mobility of many nutrients as they use them in their growth processes.
8. Microorganisms are an important part of ecosystems, or self-regulating biological communities and their physical environments. Microorganisms play an important role in succession, or the predictable changes that occur in ecosystems when they are disturbed.
9. Extreme environments restrict the range of microbial types able to survive and function. This can be due to physical factors such as temperature, pH, pressure, or salinity. Many microorganisms found in “extreme” environments are especially adapted not only to survive, but to function metabolically under these particular conditions.
10. Methods used to study microbial interactions and microbial ecology provide information on environmental characteristics; microbial biomass, numbers, types and activity; and community structure. Microscopic, chemical, enzymatic, and molecular techniques are used in these studies.
11. It is now possible to determine the nucleic acid sequences of specific microorganisms or organelles isolated from natural environments and to study the phylogeny of uncultured microorganisms. This should lead to important new advances in the study of microbial ecology.

Everything is everywhere, the environment selects.

M. W. Beijerinck

In previous chapters microorganisms usually have been considered as isolated entities. The basic characteristics of microorganisms, including the structure and function of microbial cells, metabolism, growth and the control of growth, have been discussed. In addition, metabolism, genetics, and molecular aspects of microorganisms, including genomics, have been described. In this chapter, microbial interactions with both the physical environment and with other organisms will be considered.

28.1 Foundations of Microbial Ecology

Two major themes will be developed in this chapter: (1) the nature of microbial relationships with other living organisms, or the nature of **symbioses**, and (2) the interactions of these organisms with each other and with their nonliving physical environment, or the area of **microbial ecology** (**Box 28.1**). The term symbiosis is used in its original broadest sense, as an association of two or more different species of organisms, as suggested by H. A. deBary in 1879.

Microorganisms function as **populations** or assemblages of similar organisms, and as **communities**, or mixtures of different microbial populations. These microorganisms have evolved while interacting with the inorganic world and with higher organisms, and they largely play beneficial and vital roles; disease-causing organisms are only a minor component of the microbial world. Microorganisms, as they interact with other organisms and their environment, also contribute to the functioning of **ecosystems**, or self-regulating biological communities and their physical environment (p. 622). Knowledge of these interactions is important in understanding both microbial contributions to the natural world and microbial roles in disease processes.

A major problem in understanding microbial interactions is that most microscopically observable microorganisms cannot be grown. The differences between observable and culturable microorganisms, which limit this field even today, have been noted for at least 70 years. This problem was discussed in a 1932 textbook on soil microbiology written by Selman Waksman, the discoverer of streptomycin, and has not yet been solved (*see section 6.5*). The use of molecular techniques, however, is providing valuable information on these still uncultured microorganisms, and rapid progress is being made in this area. This remains as a central challenge in attempting to understand microbial interactions, microbial ecology, and biology itself.

28.2 Microbial Interactions

Microorganisms can be physically associated with other organisms in a variety of ways. One organism can be located on the surface of another, as an **ectosymbiont**. In this case, the ectosymbiont usually is a smaller organism located on the surface of a larger organism. Often, dissimilar organisms of similar size are in physical contact. The term **consortium** can be used to describe this physical relationship. Consortia in aquatic environments are frequently complex, involving multiple layers of similar-looking microorganisms that often have complementary physiological properties. In contrast, one organism can be located within another organism as an **endosymbiont**. There also are many cases in which microorganisms live on both the inside and the outside of another organism, a phenomenon called **ecto/endosymbiosis**. Interesting examples of ecto/endosymbioses include a *Thiothrix* species, a sulfur-using bacterium, which is attached to the surface of a mayfly larva and which itself contains a parasitic bacterium. Fungi associated with plant roots (mycorrhizal fungi) often contain endosymbiotic bacteria, as well as having bacteria living on their surfaces (*see pp. 679–82*).

Box 28.1

Microbial Ecology and Environmental Microbiology

The term “microbial ecology” is now used in a general way to describe the presence and contributions of microorganisms, through their activities, to the places where they are found. Students of microbiology should be aware that much of the information on microbial presence and contributions to soils, waters, and associations with plants, now described by this term, would have been considered as “environmental microbiology” in the past. Thomas D. Brock, the discoverer of *Thermus aquaticus*, which is known the world over as the source of *Taq* polymerase for the polymerase chain reaction (PCR), has given a definition of microbial ecology that may be useful: “Microbial ecology is the study of the behavior and activities of microorganisms in their natural environments.” The important operator in this sentence is *their* environment instead of *the* environment. To emphasize this point, Brock has noted that “microbes are small; their environments also are small.” In these small environments or “microenvironments,” other kinds of microorganisms (and macroorganisms) often also are present, a critical point that was emphasized by Sergei Winogradsky in 1947.

Environmental microbiology, in comparison, relates primarily to all-over microbial processes that occur in a soil, water, or food, as examples. It is not concerned with the particular “microenvironment” where the microorganisms actually are functioning, but with the broader-scale effects of microbial presence and activities. One can study these microbially mediated processes and their possible global impacts at the scale of “environmental microbiology” without knowing about the specific microenvironment (and the organisms functioning there) where these processes actually take place. However, it is critical to be aware that microbes function in their localized environments and affect ecosystems at greater scales, including causing global-level effects. In the last decades the term “microbial ecology” largely has lost its original meaning, and recently the statement has been made that “microbial ecology has become a ‘catch-all’ term.” As you read various textbooks and scientific papers, possible differences between “microbial ecology” and “environmental microbiology” should be kept in mind.

These physical associations can be intermittent and cyclic or permanent. Examples of intermittent and cyclic associations of microorganisms with plants and marine animals are shown in **table 28.1**. Important human diseases, including listeriosis, malaria, leptospirosis, legionellosis, and vaginosis also involve such intermittent and cyclic symbioses. These diseases will be discussed in chapters 39 and 40. Interesting permanent relationships also occur between bacteria and animals, as shown in **table 28.2**. Hosts include squid, leeches, aphids, nematodes, and mollusks. In each of these cases, an important characteristic of the host animal is conferred by the permanent bacterial symbiont.

Although it is possible to observe microorganisms in these varied physical associations with other organisms, the fact that there is some type of physical contact provides no information on the types of interactions that might be occurring. These interactions can be positive (mutualism, protocooperation, and

Table 28.1 Intermittent and Cyclical Symbioses of Microorganisms with Plants and Marine Animals

Symbiosis	Host	Cyclical Symbiont
Plant-bacterial	<i>Gunnera</i> (tropical angiosperm)	<i>Nostoc</i> (cyanobacterium)
	<i>Azolla</i> (rice paddy fern)	<i>Anabaena</i> (cyanobacterium)
	<i>Phaseolus</i> (bean)	<i>Rhizobium</i> (N ₂ fixer)
	<i>Ardisia</i> (angiosperm)	<i>Protobacterium</i>
Marine animals	Coral coelenterates	<i>Symbiodinium</i> (dinomastigote)
	Luminous fish	<i>Vibrio</i> , <i>Photobacterium</i>
	Squid	<i>Photobacterium fischerei</i>

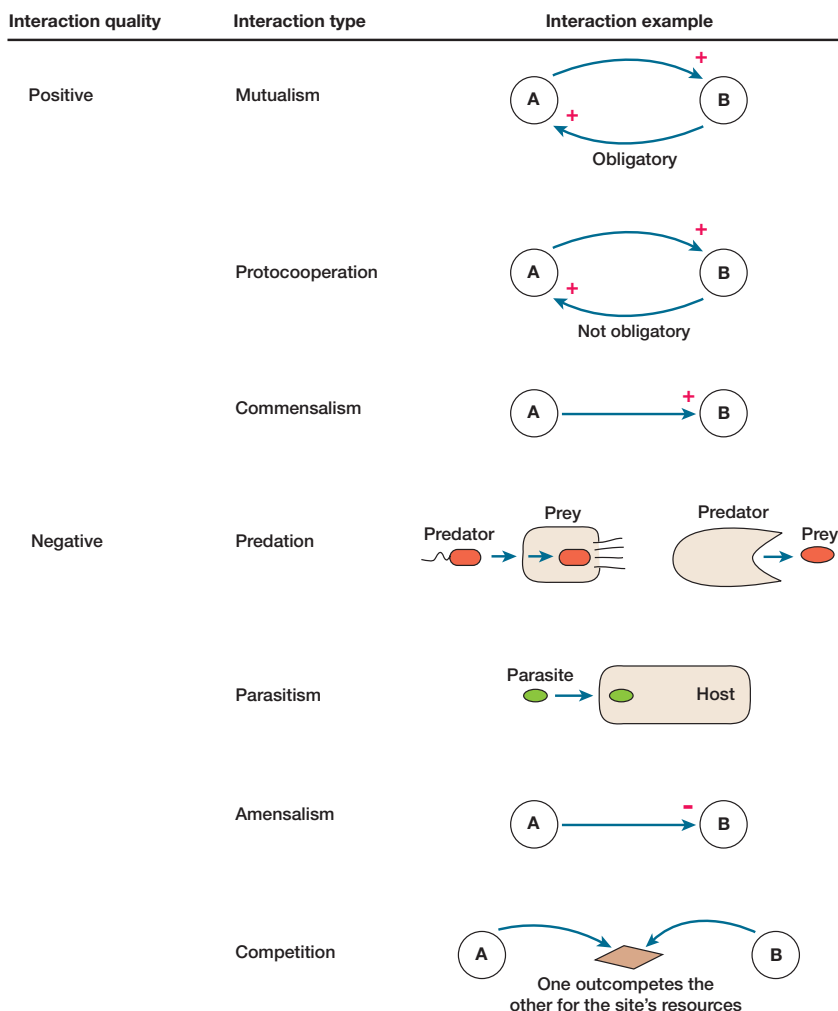
Adapted from L. Margulis and M. J. Chapman. 1998. Endosymbioses: Cyclical and permanent in evolution. *Trends in Microbiology* 6(9):342–46, tables 1, 2, and 3.

Table 28.2 Examples of Permanent Bacterial-Animal Symbioses and the Characteristics Contributed by the Bacterium to the Symbiosis

Animal Host	Symbiont	Symbiont Contribution
Sepiolid squid (<i>Euprymna scolopes</i>)	Luminous bacterium	Luminescence (<i>Vibrio fischeri</i>)
Medicinal leech (<i>Hirudo medicinalis</i>)	Enteric bacterium (<i>Aeromonas veronii</i>)	Blood digestion
Aphid (<i>Schizaphis graminum</i>)	Bacterium (<i>Buchnera aphidicola</i>)	Amino acid synthesis
Nematode worm (<i>Heterorhabditis</i> spp.)	Luminous bacterium (<i>Photorhabdus luminescens</i>)	Predation and antibiotic synthesis
Shipworm mollusk (<i>Lyrodus pedicellatus</i>)	Gill cell bacterium	Cellulose digestion and nitrogen fixation

Source: From E. G. Ruby, 1999. Ecology of a benign “infection”: Colonization of the squid luminous organ by *Vibrio fischeri*. In *Microbial ecology and infectious disease*, E. Rosenberg, editor, American Society for Microbiology, Washington, D.C., 217–31, table 1.

Figure 28.1 Microbial Interactions. Basic characteristics of positive (+) and negative (–) interactions that can occur between different organisms.



commensalism) or negative (predation, parasitism, amensalism, and competition) as shown in **figure 28.1**. These interactions will be discussed next.

1. Define the terms symbiosis and microbial ecology. How are they similar and different?
2. In what ways can different microorganisms be in physical contact?
3. Define the terms population, community, and ecosystem.
4. List several important diseases that involve cyclic and intermittent symbioses.

Mutualism

Mutualism [Latin *mutuus*, borrowed or reciprocal] defines the relationship in which some reciprocal benefit accrues to both partners. This is an obligatory relationship in which the **mutualist** and the host are metabolically dependent on each other. Several examples of mutualism are presented next.

The protozoan-termite relationship is a classic example of mutualism in which the flagellated protozoa live in the gut of termites and wood roaches. (**figure 28.2a**). These flagellates exist on a diet of carbohydrates, acquired as cellulose ingested by their host (**figure 28.2b**). The protozoa engulf wood particles, digest the cellulose, and metabolize it to acetate and other products. Termites oxidize the acetate released by their flagellates. Because the host is almost always incapable of synthesizing cellulases (enzymes that catalyze the hydrolysis of cellulose), it is dependent on the mutualistic protozoa for its existence.

This mutualistic relationship can be readily tested in the laboratory if wood roaches are placed in a bell jar containing wood chips and a high concentration of O_2 . Because O_2 is toxic to the flagellates, they die. The wood roaches are unaffected by the high O_2 concentration and continue to ingest wood, but they soon die of starvation due to a lack of cellulases.

Lichens are another excellent example of mutualism (**figure 28.3**). **Lichens** are the association between specific ascomycetes (the fungus) and certain genera of either green algae or cyanobacteria. In a lichen, the fungal partner is termed the **mycobiont** and the algal or cyanobacterial partner, the **phycobiont**.



(a)



(b)

Figure 28.2 Mutualism. Light micrographs of (a) a worker termite of the genus *Reticulitermes* eating wood ($\times 10$), and (b) *Trichonympha*, a multiflagellated protozoan from the termite's gut ($\times 135$). Notice the many flagella that occur over most of its length. The ability of *Trichonympha* to break down cellulose enables termites to use wood as a food source.

The remarkable aspect of this mutualistic association is that its morphology and metabolic relationships are so constant that lichens are assigned generic and species names. The characteristic morphology of a given lichen is a property of the mutualistic association and is not exhibited by either symbiont individually. *Ascomycetes* (pp. 560–61); *cyanobacteria* (pp. 471–76)

Because the phycobiont is a photoautotroph—dependent only on light, carbon dioxide, and certain mineral nutrients—the fungus can get its organic carbon directly from the alga or cyanobacterium. The fungus often obtains nutrients from its partner by haustoria (projections of fungal hyphae) that penetrate the phycobiont cell wall. It also uses the O_2 produced during phycobiont photophosphorylation in carrying out respiration. In turn the fungus protects the phycobiont from excess light intensities, provides water and minerals to it, and creates a firm substratum within which the phycobiont can grow protected from environmental stress.

Many marine invertebrates (sponges, jellyfish, sea anemones, corals, ciliates) harbor endosymbiotic, spherical algal cells called **zooxanthellae** within their tissue (**figure 28.4a**). Because the degree of host dependency on the mutualistic alga is somewhat variable, only one well-known example is presented.

The hermatypic (reef-building) corals (**figure 28.4b**) satisfy most of their energy requirements using their zooxanthellae. Pigments produced by the coral protect the algae from the harmful effects of ultraviolet radiation. Clearly the zooxanthellae also benefit the coral because the calcification rate is at least 10 times greater in the light than in the dark. Hermatypic corals lacking zooxanthellae have a very low rate of calcification. Based on their stable carbon isotopic composition, it has been determined that most of the organic carbon in the tissues of the hermatypic corals has come from the zooxanthellae. Because of this coral-algal mutualistic relationship—capturing, conserving, and cycling nutrients and energy—coral reefs are among the most productive and successful of known ecosystems.



Figure 28.3 Lichens. Crustose (encrusting) lichens growing on a granite post.

Sulfide-Based Mutualisms

Tube worm–bacterial relationships exist several thousand meters below the surface of the ocean, where the Earth's crustal plates are spreading apart (**figure 28.5**). Vent fluids are anoxic, contain high concentrations of hydrogen sulfide, and can reach a temperature of 350°C . The seawater surrounding these vents has sulfide

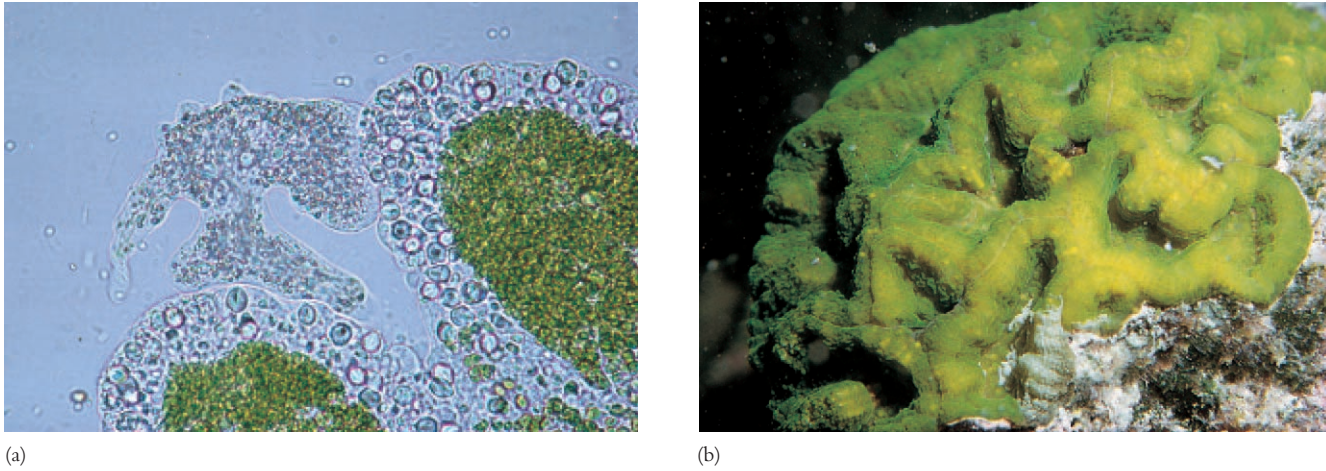


Figure 28.4 Zooxanthellae. (a) Zooxanthellae (green) within the tip of a hydra tentacle ($\times 150$). (b) The green color of this rose coral (*Manilina*) is due to the abundant zooxanthellae within its tissues.

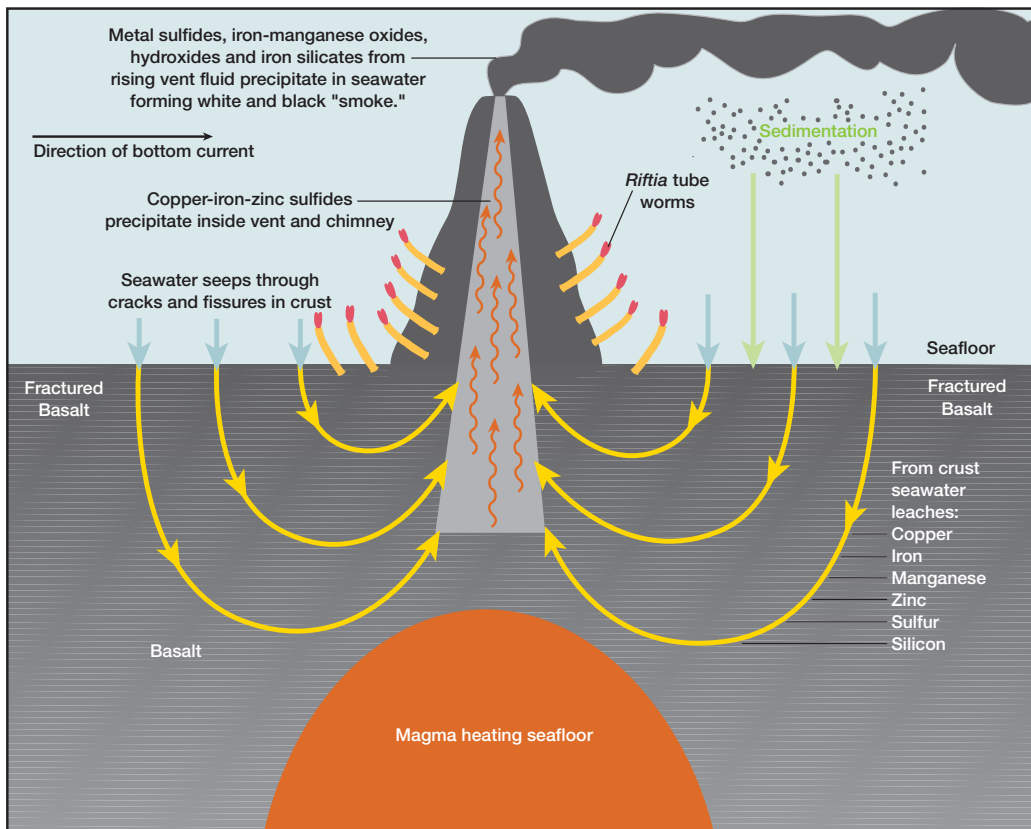


Figure 28.5 Basic Structure of a Hydrothermal Vent with its Mutualistic Microbe-Animal Associations. Reduced chemicals including sulfide are released as seawater penetrates the fractured basaltic ocean floor, is heated, and returns as vent fluid to the ocean, creating environments for growth of the tube worms and their procaryotic mutualists.

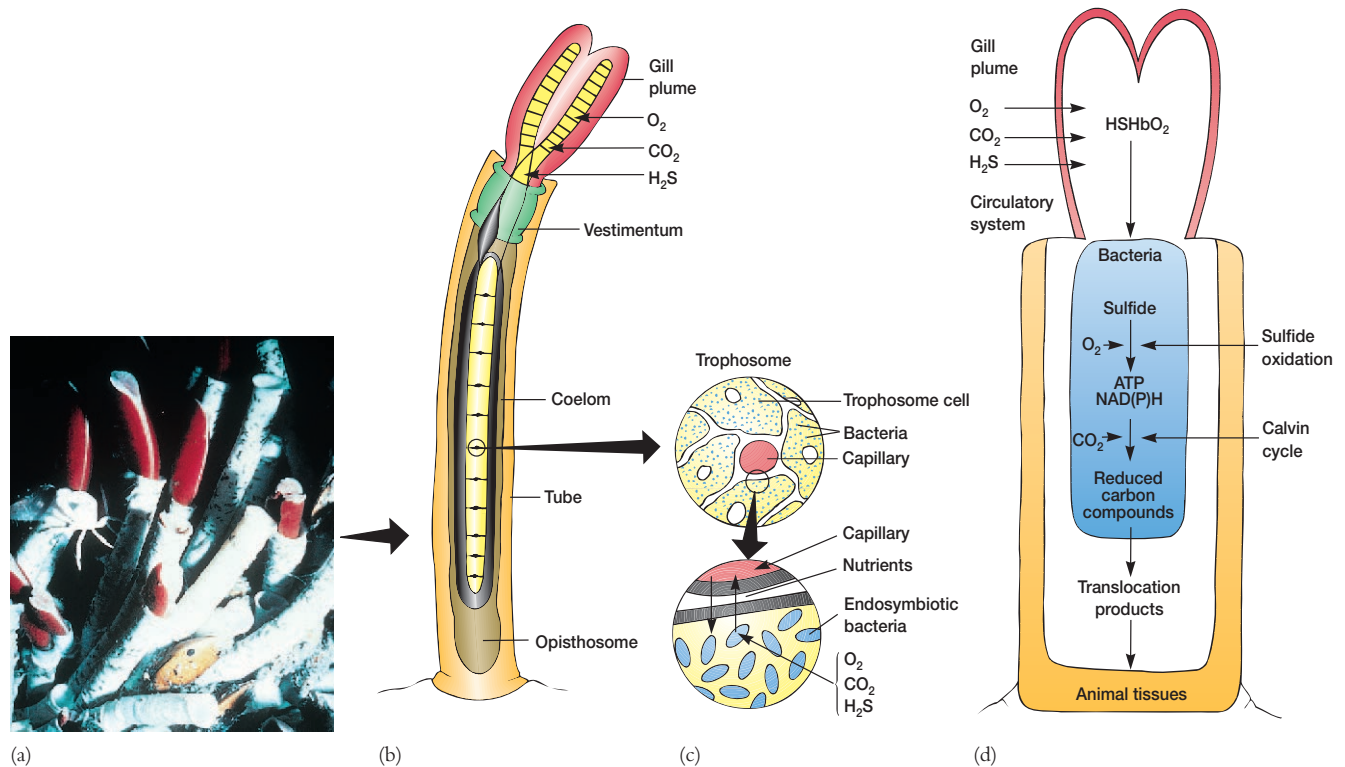


Figure 28.6 The Tube Worm—Bacterial Relationship. (a) A community of tube worms (*Riftia pachyptila*) at the Galápagos Rift hydrothermal vent site (depth 2,550 m). Each worm is more than a meter in length and has a 20 cm gill plume. (b, c) Schematic illustration of the anatomical and physiological organization of the tube worm. The animal is anchored inside its protective tube by the vestimentum. At its anterior end is a respiratory gill plume. Inside the trunk of the worm is a trophosome consisting primarily of endosymbiotic bacteria, associated cells, and blood vessels. At the posterior end of the animal is the opisthosome, which anchors the worm in its tube. (d) Oxygen, carbon dioxide, and hydrogen sulfide are absorbed through the gill plume and transported to the blood cells of the trophosome. Hydrogen sulfide is bound to the worm's hemoglobin (HSHbO₂) and carried to the endosymbiont bacteria. The bacteria oxidize the hydrogen sulfide and use some of the released energy to fix CO₂ in the Calvin cycle. Some fraction of the reduced carbon compounds synthesized by the endosymbiont is translocated to the animal's tissues.

concentrations around 250 μM and temperatures 10 to 20°C above the normal seawater temperature of 2.1°C.

The giant (>1 m in length), red, gutless tube worms (*Riftia* spp.) near these hydrothermal vents provide an example of a unique form of mutualism and animal nutrition in which chemolithotrophic bacterial endosymbionts are maintained within specialized cells of the tube worm host (figure 28.6). To date all attempts to culture these microorganisms have been unsuccessful.

The tube worm takes up hydrogen sulfide from the seawater and binds it to hemoglobin (the reason the worms are bright red). The hydrogen sulfide is then transported in this form to the bacteria, which use the sulfide-reducing power to fix carbon dioxide in the Calvin cycle (see figure 10.4). The CO₂ required for this cycle is transported to the bacteria in three ways: freely dissolved in the blood, bound to hemoglobin, and in the form of organic acids such as malate and succinate. These acids are decarboxylated to release CO₂ in the trophosome, the tissue containing bacterial symbionts. Using these mechanisms, the bacteria syn-

thesize reduced organic material from inorganic substances. The organic material is then supplied to the tube worm through its circulatory system and serves as the main nutritional source for the tissue cells.

Methane-Based Mutualisms

Other unique food chains involve methane-fixing microorganisms as the first step in providing organic matter for consumers. Methanotrophs, bacteria capable of using methane, occur as intracellular symbionts of methane-vent mussels. In these mussels the thick fleshy gills are filled with bacteria. In addition, methanotrophic carnivorous sponges have been discovered in a mud volcano at a depth of 4,943 m in the Barbados Trench. Abundant methanotrophic symbionts were confirmed by the presence of enzymes related to methane oxidation in sponge tissues. These sponges are not satisfied to use bacteria to support themselves; they also trap swimming prey to give variety to their diet.

Microorganism-Insect Mutualisms

Mutualistic associations are common in the insects. This is related to the foods used by insects, which often include plant sap or animal fluids lacking in essential vitamins and amino acids. The required vitamins and amino acids are provided by bacterial symbionts in exchange for a secure physical habitat and ample nutrients. The aphid is an excellent example of this mutualistic relationship. This insect contains *Buchnera aphidicola* in its cytoplasm, and a mature insect contains literally millions of these bacteria in its body. The *Buchnera* provides its host with amino acids, particularly tryptophan, and if the insect is treated with antibiotics, it dies. *Wolbachia pipientis*, a rickettsia, is a cytoplasmic endosymbiont found in 15 to 20% of insect species and can control the reproduction of its host. This microbial association is thought to be a major factor in the evolution of sex and speciation in the parasitic wasps. *Wolbachia* also can cause cytoplasmic incompatibility in insects, parthenogenesis in butterflies, and the feminization of genetic males in isopods. What could be the advantage to the *Wolbachia*? By limiting sexual variability, the bacterium might benefit by creating a more stable asexual environment for its own longer-term maintenance. Our understanding of microbe-insect mutualisms, including the role of *Wolbachia* in insects, is constantly expanding with the increased use of molecular techniques.

1. What is a lichen? Discuss the benefits the phycobiont and mycobiont provide each other.
2. What is the critical characteristic of a mutualistic relationship?
3. How do tube worms obtain energy and organic compounds for their growth?
4. What is the source of the waters released in a deep hydrothermal vent, and how is it heated?
5. What are important roles of bacteria, such as *Buchnera* and *Wolbachia*, in insects?

The Rumen Ecosystem

Ruminants are a group of herbivorous animals that have a stomach divided into four compartments and chew a cud consisting of regurgitated, partially digested food. Examples include cattle, deer, elk, camels, buffalo, sheep, goats, and giraffes. This feeding method has evolved in animals that need to eat large amounts of food quickly, chewing being done later at a more comfortable or safer location. More importantly, by using microorganisms to degrade the thick cellulose walls of grass and other vegetation, ruminants digest vast amounts of otherwise unavailable forage. Because ruminants cannot synthesize cellulases, they have established a mutualistic relationship with anaerobic microorganisms that produce these enzymes. Cellulases hydrolyze the β (1 $\rightarrow$ 4) linkages between successive D-glucose residues of cellulose and release glucose, which is then fermented to organic acids such as acetate, butyrate, and propionate (see figure 9.10). These organic acids are the true energy source for the ruminant.

The upper portion of a ruminant's stomach expands to form a large pouch called the **rumen** (figure 28.7) and also a smaller

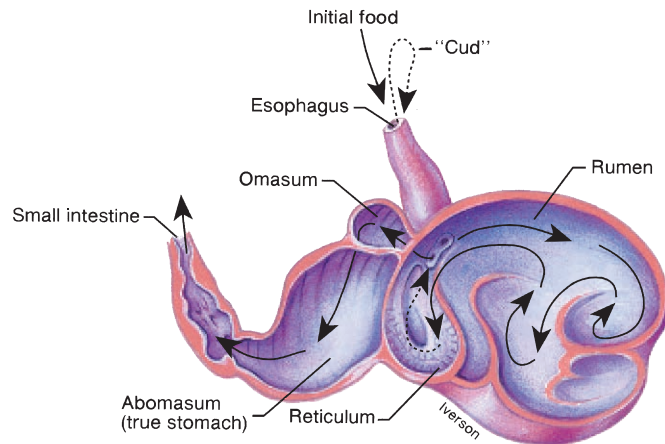


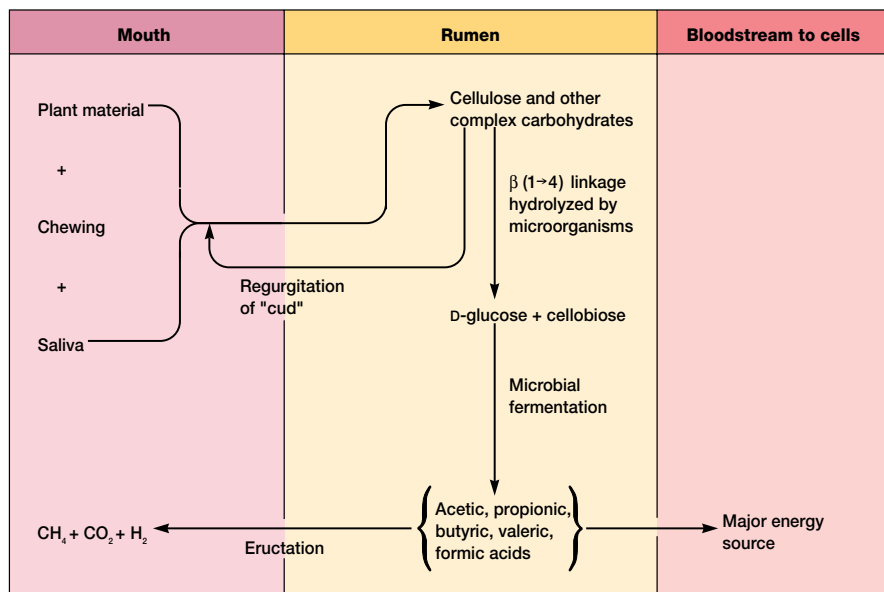
Figure 28.7 Ruminant Stomach. The stomach compartments of a cow. The microorganisms are active mainly in the rumen. Arrows indicate direction of food movement.

honeycomblke reticulum. The bottom portion of the stomach consists of an antechamber called the omasum, with the “true” stomach (abomasum) behind it.

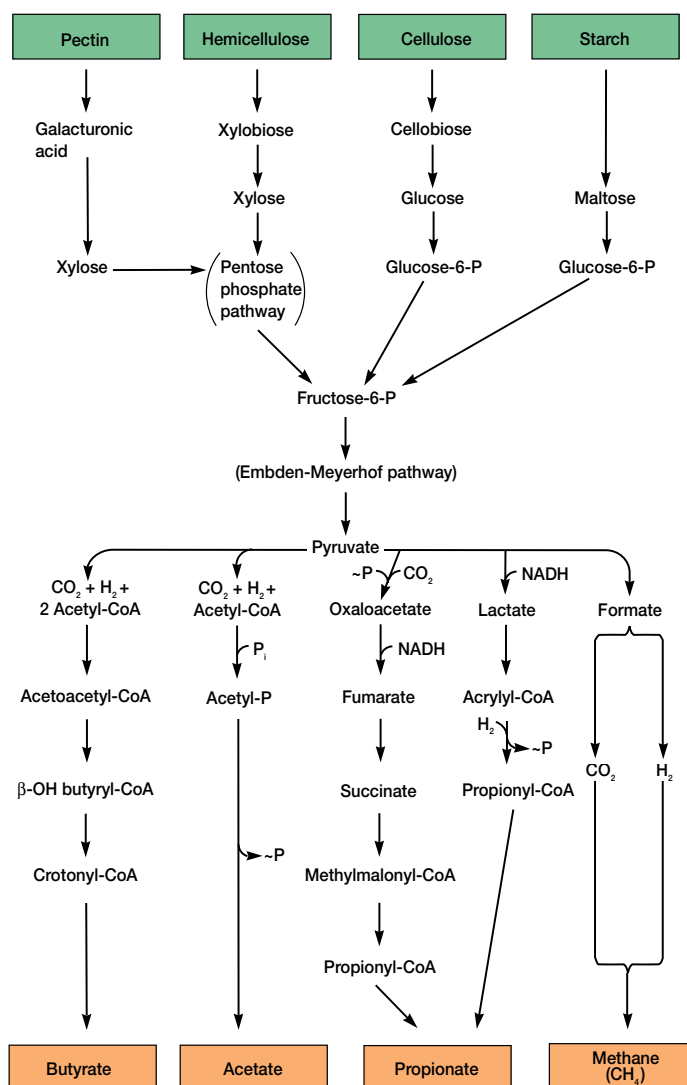
The insoluble polysaccharides and cellulose eaten by the ruminant are mixed with saliva and enter the rumen. Within the rumen, food is churned in a constant rotary motion and eventually reduced to a pulpy mass, which is partially digested and fermented by microorganisms. Later the food moves into the reticulum. It is then regurgitated as a “cud,” which is thoroughly chewed for the first time. The food is mixed with saliva, reswallowed, and reenters the rumen while another cud is passed up to the mouth. As this process continues, the partially digested plant material becomes more liquid in nature. The liquid then begins to flow out of the reticulum and into the lower parts of the stomach: first the omasum and then the abomasum. It is in the abomasum that the food encounters the host's normal digestive enzymes and the digestive process continues in the regular mammalian way.

The rumen contains a large and diverse microbial community (about 10^{12} organisms per milliliter), including procaryotes, anaerobic fungi such as *Neocallimastix*, and ciliates and other protozoans. Food entering the rumen is quickly attacked by the cellulolytic anaerobic procaryotes, fungi, and protozoa. Although the masses of procaryotes and protozoa are approximately equal, the processing of rumen contents is carried out mainly by the procaryotes. Microorganisms break down the plant material, as illustrated in figure 28.8. Because the reduction potential in the rumen is -30 mV, all indigenous microorganisms engage in anaerobic metabolism. The bacteria ferment carbohydrates to fatty acids, carbon dioxide, and hydrogen. The archaea (methanogens) produce methane (CH_4) from acetate, CO_2 , and H_2 .

Dietary carbohydrates degraded in the rumen include soluble sugars, starch, pectin, hemicellulose, and cellulose. The largest percentage of each carbohydrate is fermented to volatile fatty acids (acetic, propionic, butyric, formic, and valeric), CO_2 , H_2 , and methane. Fatty acids produced by the rumen organisms are absorbed into the bloodstream and are oxidized by the animal as its



(a)



(b)

Figure 28.8 Rumen Biochemistry.

(a) An overview of the biochemical-physiological processes occurring in various parts of a cow's digestive system. (b) More specific biochemical pathways involved in rumen fermentation of the major plant carbohydrates. The top boxes represent substrates and the bottom boxes some of the end products.

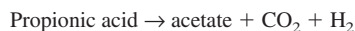
main source of energy. The CO_2 and methane, produced at a rate of 200 to 400 liters per day in a cow, are released by eructation [Latin *eructare*, to belch], a continuous, scarcely audible reflex process similar to belching. ATP produced during fermentation is used to support the growth of rumen microorganisms. These microorganisms in turn produce most of the vitamins needed by the ruminant. In the remaining two stomachs, the microorganisms, having performed their symbiotic task, are digested to yield amino acids, sugars, and other nutrients for ruminant use.

Syntrophism

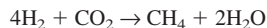
Syntrophism [Greek *syn*, together, and *trophe*, nourishment] is an association in which the growth of one organism either depends on or is improved by growth factors, nutrients, or substrates provided by another organism growing nearby. Sometimes both organisms benefit. This type of mutualism is also known as cross-feeding or the satellite phenomenon.

A very important syntrophism occurs in anaerobic methanogenic ecosystems such as sludge digesters (see section 29.6), anaerobic freshwater aquatic sediments, and flooded soils. In these environments, fatty acids can be degraded to produce H_2 and methane by the interaction of two different bacterial groups. Methane production by methanogens depends on **interspecies hydrogen transfer**. A fermentative bacterium generates hydrogen gas, and the methanogen uses it quickly as a substrate for methane gas production.

Various fermentative bacteria produce low molecular weight fatty acids that can be degraded by anaerobic bacteria such as *Syntrophobacter* to produce H_2 as follows:



Syntrophobacter uses protons ($\text{H}^+ + \text{H}^+ \rightarrow \text{H}_2$) as terminal electron acceptors in ATP synthesis. The bacterium gains sufficient energy for growth only when the H_2 it generates is consumed. The products H_2 and CO_2 are used by methanogenic archaea (e.g., *Methanospirillum*) as follows:



By synthesizing methane, *Methanospirillum* maintains a low H_2 concentration in the immediate environment of both bacteria. Continuous removal of H_2 promotes further fatty acid fermentation and H_2 production. If the hydrogen is not consumed, it will inhibit *Syntrophobacter*. Because increased H_2 production and consumption stimulate the growth rates of *Syntrophobacter* and *Methanospirillum*, both participants in the relationship benefit.

1. What structural features of the rumen make it suitable for a herbivorous type of diet? Why does a cow chew its cud?
2. What biochemical roles do the rumen microorganisms play in this type of symbiosis?
3. What is syntrophism? Is physical contact required for this relationship?
4. What is interspecies hydrogen transfer, and why can this be beneficial to both the producer and consumer of hydrogen?

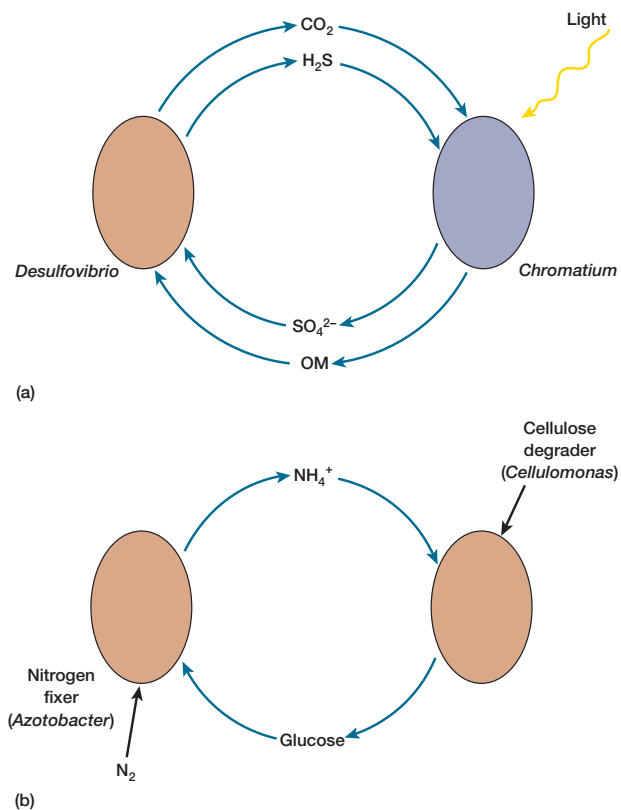


Figure 28.9 Examples of Protocooperative Symbiotic Processes. (a) The organic matter (OM) and sulfate required by *Desulfovibrio* are produced by the *Chromatium* in its photosynthesis-driven reduction of CO_2 to organic matter and oxidation of sulfide to sulfate. (b) *Azotobacter* uses glucose provided by a cellulose-degrading microorganism such as *Cellulomonas*, which uses the nitrogen fixed by *Azotobacter*.

Protocooperation

As noted in figure 28.1, **protocooperation** is a mutually beneficial relationship, similar to that which occurs in mutualism, but in protocooperation, this relationship is not obligatory. As noted in this figure, beneficial complementary resources are provided by each of the paired microorganisms. The organisms involved in this type of relationship can be separated, and if the resources provided by the complementary microorganism are supplied in the growth environment, each microorganism will function independently. Two examples of this type of relationship are the association of *Desulfovibrio* and *Chromatium* (figure 28.9a), in which the carbon and sulfur cycles are linked, and the interaction of a nitrogen-fixing microorganism with a cellulolytic organism such as *Cellulomonas* (figure 28.9b). In the second example, the cellulose-degrading microorganism liberates glucose from the cellulose, which can be used by nitrogen-fixing microbes.

An excellent example of a protocooperative biodegradative association is shown in figure 28.10. In this case 3-chlorobenzoate degradation depends on the functioning of microorganisms with complementary capabilities. If any one of the three microorgan-

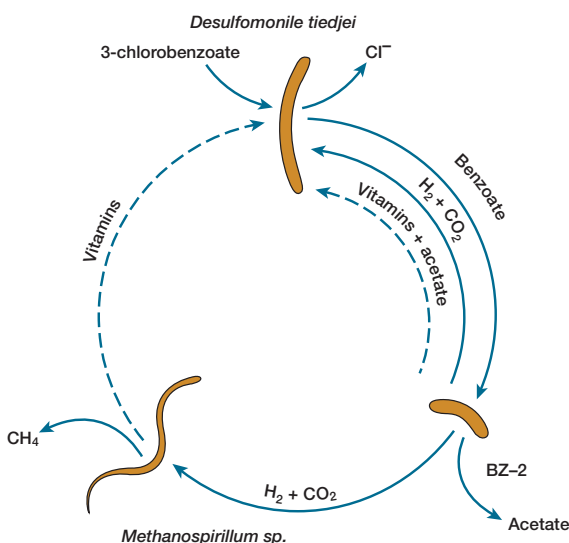


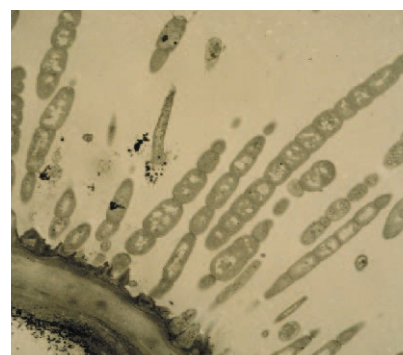
Figure 28.10 Associations in a Defined Three-Membered Protocooperative and Commensalistic Community That Can Degrade 3-Chlorobenzoate. If any member is missing, degradation will not take place. The solid arrows demonstrate nutrient flows, and the dashed lines represent hypothesized flows.



Figure 28.11 A Marine Worm-Bacterial Protocooperative Relationship. *Alvinella pompejana*, a 10 cm long worm, forms a protocooperative relationship with bacteria that grow as long threads on the worm's surface. These waters contain sulfur compounds that are used by the bacteria as electron acceptors in the presence of fumarate and pyruvate, and the worm uses the bacteria as a food source. The bacteria and *Alvinella* are found in tunnels near the black smoker-heated water fonts.



(a)



(b)

Figure 28.12 A Marine Crustacean-Bacterial Protocooperative Relationship. (a) A picture of the marine shrimp *Rimicaris exoculata* clustering around a hydrothermal vent area, showing the massive development of these crustaceans in the area where chemolithotrophic bacteria grow using sulfide as an electron and energy source. The bacteria, which grow on the vent openings and also on the surface of the crustaceans, fix carbon from CO_2 in their autotrophic metabolism, and serve as the nutrient for the shrimp. (b) An electron micrograph of a thin section across the leg of the marine crustacean *Rimicaris exoculata*, showing the chemolithotrophic bacteria that cover the surface of the shrimp. The filamentous nature of these bacteria, upon which this commensalistic relationship is based, is evident in this thin section.

isms is not present and active, the degradation of the substrate will not occur.

In other protocooperative relations, sulfide-dependent autotrophic filamentous microorganisms fix carbon dioxide and synthesize organic matter that serves as a carbon and energy source for a heterotrophic organism. One of the most interesting such relationship is the Pompeii worm (*Alvinella pompejana*), named for the

deep-sea submersible from Woods Hole, Massachusetts. This unusual organism is 10 cm in length and lives in tunnels near waters that approach 80°C in a deep area of the Pacific Ocean (figure 28.11). It uses as a nutrient source bacteria that appear to oxidize organic matter and reduce sulfur compounds. A deep-sea crustacean has been discovered that uses sulfur-oxidizing autotrophic bacteria as its food source. This shrimp, *Rimicaris exoculata* (figure 28.12)

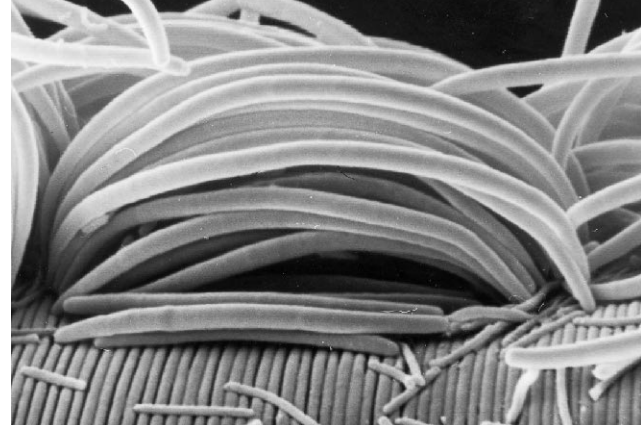
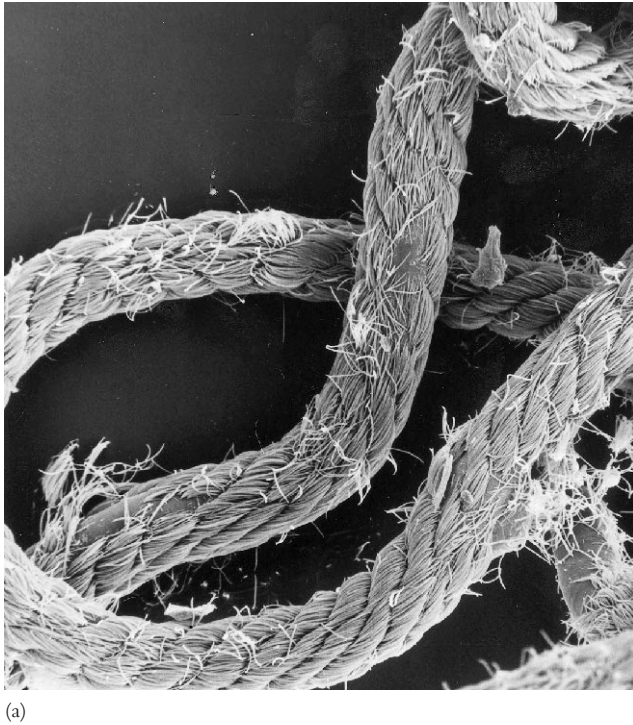


Figure 28.13 A Marine Nematode-Bacterial Protocooperative Relationship. Marine free-living nematodes, which grow at the oxidized-reduced zone where sulfide and oxygen are present, are covered by sulfide-oxidizing bacteria. The bacteria protect the nematode by decreasing sulfide concentrations near the worm, and the worm uses the bacteria as a food source. (a) The marine nematode *Eubostrichus parasitiferus* with bacteria arranged in a characteristic helix pattern. Bar scale = 100 μm . (b) The chemolithotrophic bacteria attached to the cuticle of the marine nematode *Eubostrichus parasitiferus*. Cells are fixed to the nematode surface at both ends. Bar = 10 μm . See also figure 28.17.

has filamentous sulfur-oxidizing bacteria growing on its surface. When these are dislodged the shrimp ingests them. This nominally “blind” shrimp can respond to the glow emitted by the black smoker, using a reflective organ on its back. The organ is sensitive to a light wavelength that is not detectable by humans.

Another interesting example of bacterial epigrowth is shown by nematodes, including *Eubostrichus parasitiferus*, that live at the interface between aerobic and anaerobic sulfide-containing marine sediments (figure 28.13a). These animals are covered by sulfide-oxidizing bacteria that are present in intricate patterns (figure 28.13b). The bacteria not only decrease levels of toxic sulfide, which often surround the nematodes, but they also serve as a food supply.

In 1990, hydrothermal vents were discovered in a freshwater environment, at the bottom of Lake Baikal, the oldest (25 million years old) and deepest lake in the world. This lake is located in the far east of Russia (figure 28.14a) and has the largest volume of any freshwater lake (not the largest area—which is Lake Superior). The bacterial growths, with long white strands, are in the center of the vent field where the highest temperatures are found (figure 28.14b). At the edge of the vent field, where the water temperature is lower, the bacterial mat ends, and sponges, gastropods, and other organisms, which use the sulfur-oxidizing bacteria as a food source, are present (figure 28.14c). Similar although less developed areas have been found in Yellowstone Lake, Wyoming.

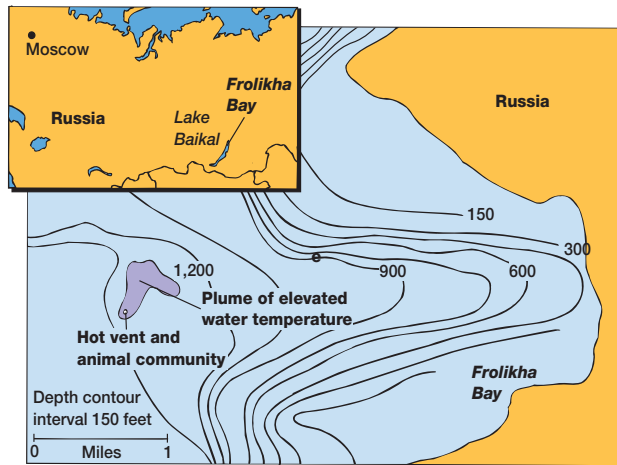
A hydrogen sulfide-based ecosystem has been discovered in southern Romania that is closer to the earth’s surface. Caves in the area contain mats of microorganisms that fix carbon dioxide using hydrogen sulfide as the reductant. Forty-eight species of cave-adapted invertebrates are sustained by this chemoautotrophic base.

A form of protocooperation also occurs when a population of similar microorganisms monitors its own density, the process of quorum sensing, which was discussed in section 6.5. The microorganisms produce specific autoinducer compounds, and as the population increases and the concentration of these compounds reaches critical levels, specific genes are expressed. These responses are important for microorganisms that form associations with plants and animals, and particularly for human pathogens.

1. Why are *Alvinella*, *Rimicaris* and *Eubostrichus* good examples of protocooperative microorganism-animal interactions?
2. What important freshwater hydrothermal vent communities have been described?

Commensalism

Commensalism [Latin *com*, together, and *mensa*, table] is a relationship in which one symbiont, the **commensal**, benefits while the other (sometimes called the host) is neither harmed nor helped as shown in figure 28.1. This is a unidirectional process. Often both the host and the commensal “eat at the same table.” The spatial proximity of the two partners permits the commensal to feed on substances captured or ingested by the host, and the commensal often obtains shelter by living either on or in the host. The commensal is not directly dependent on the host metabolically and causes it no particular harm. When the commensal is separated from its host experimentally, it can survive without being provided some factor or factors of host origin.



(a)



(b)



(c)

Figure 28.14 Hydrothermal Vent Ecosystems in Freshwater Environments. Lake Baikal (in Russia) has been found to have low temperature hydrothermal vents. (a) Location of Lake Baikal, site of the hydrothermal vent field. (b) Bacterial mat near the center of the vent field. (c) Bacterial filaments and sponges at the edge of the vent field. (a) Source: Data from the National Geographic Society.

Commensalistic relationships between microorganisms include situations in which the waste product of one microorganism is the substrate for another species. An example is nitrification, the oxidation of ammonium ion to nitrite by microorganisms such as *Nitrosomonas*, and the subsequent oxidation of the nitrite to nitrate by *Nitrobacter* and similar bacteria (see pp. 193–94). *Nitrobacter* benefits from its association with *Nitrosomonas* because it uses nitrite to obtain energy for growth.

Commensalistic associations also occur when one microbial group modifies the environment to make it more suited for another organism. For example, in the intestine the common, non-pathogenic strain of *Escherichia coli* lives in the human colon, but also grows quite well outside the host, and thus is a typical commensal. When oxygen is used up by the facultatively anaerobic *E. coli*, obligate anaerobes such as *Bacteroides* are able to grow in the colon. The anaerobes benefit from their association with the host and *E. coli*, but *E. coli* derives no obvious benefit from the anaerobes. In this case the commensal *E. coli* contributes to the welfare of other symbionts. Commensalism can involve other environmental modifications. The synthesis of acidic waste products during fermentation stimulate the proliferation of more acid-tolerant microorganisms, which are only a minor part of the microbial community at neutral pHs. A good example is the succession of microorganisms during milk spoilage. When biofilms are formed (section 28.4), the colonization of a newly exposed surface by one type of microorganism (an initial colonizer) makes it possible for other microorganisms to attach to the microbially modified surface.

Commensalism also is important in the colonization of the human body and the surfaces of other animals and plants. The microorganisms associated with an animal skin and body orifices can use volatile, soluble, and particulate organic compounds from the host as nutrients (see section 31.2). Under most conditions these microbes do not cause harm, other than possibly contributing to body odor. Sometimes when the host organism is stressed or the skin is punctured, these normally commensal microorganisms may become pathogenic. These interactions will be discussed in chapter 31.

1. How does commensalism differ from protocooperation?
2. Why is nitrification a good example of a commensalistic process?
3. Why are commensalistic microorganisms important for humans? Where are these found in relation to the human body?

Predation

Predation is a widespread phenomenon where the predator engulfs or attacks the prey, as shown in figure 28.1. The prey can be larger or smaller than the predator, and this normally results in the death of the prey.

An interesting array of predatory bacteria are active in nature. Several of the best examples are shown in figure 28.15, including *Bdellovibrio*, *Vampirococcus*, and *Daptobacter*. Each of these has a unique mode of attack against a susceptible bacterium. *Bdellovibrio* penetrates the cell wall and multiplies between the wall and the

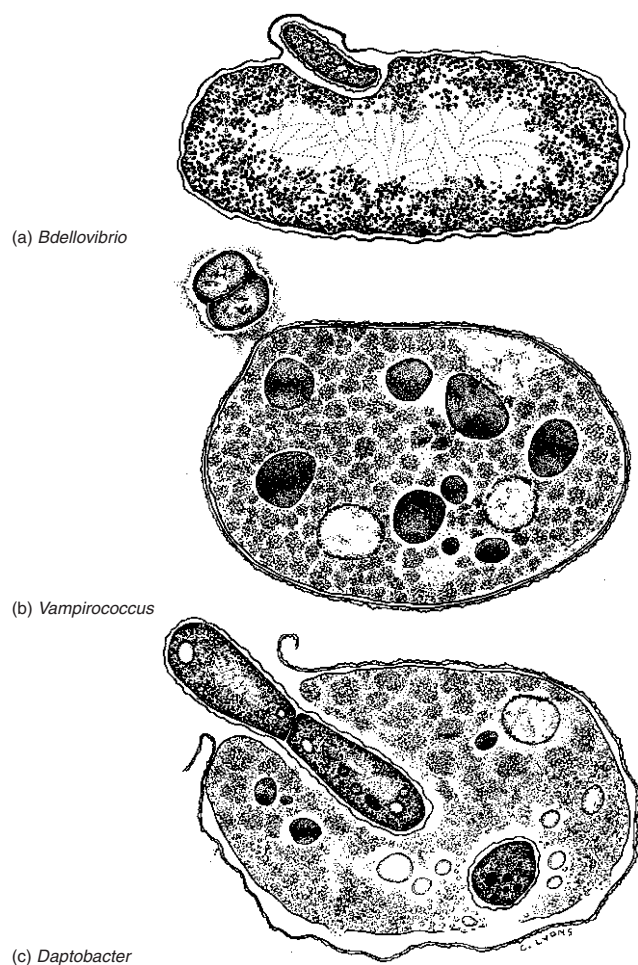


Figure 28.15 Examples of Predatory Bacteria Found in Nature. (a) *Bdellovibrio*, a periplasmic predator that penetrates the cell wall and grows outside the plasma membrane, (b) *Vampirococcus* with its unique epibiotic mode of attacking a prey bacterium, and (c) *Daptobacter* showing its cytoplasmic location as it attacks a susceptible bacterium.

plasma membrane, a periplasmic mode of attack, followed by lysis of the prey and release of progeny (see figure 22.33). This bacterium has an interesting life cycle, which is discussed in section 22.4.

Nonlytic forms also are observed. *Vampirococcus* attaches to the surface of the prey (an epibiotic relationship) and then secretes enzymes to release the cell contents. *Daptobacter* penetrates a susceptible host and uses the cytoplasmic contents as a nutrient source.

Ciliates are excellent examples of predators that engulf their prey, and based on work with fluorescently marked prey bacteria, a single ciliate can ingest 60 to 70 prey bacteria per hour! Predation on bacteria is important in the aquatic environment and in sewage treatment where the ciliates remove suspended bacteria that have not settled.

A surprising finding is that predation has many beneficial effects, especially when one considers interactive populations of predators and prey, as summarized in table 28.3. Simple ingestion and assimilation of a prey bacterium can lead to increased rates of nutrient cycling, critical for the functioning of the **microbial loop** (see section 29.1 and figure 29.4). In this process, organic matter produced through photosynthetic and chemotrophic activity is mineralized before it reaches the higher consumers, allowing the minerals to be made available to the primary producers, in a “loop.” This is important in freshwater, marine, and terrestrial environments. Ingestion and short-term retention of bacteria also is critical for functioning of ciliates in the rumen, where methanogenic bacteria contribute to the health of the ciliates by decreasing toxic hydrogen levels through using H_2 to produce methane, which then is passed from the rumen.

Predation also can provide a protective, high-nutrient environment for particular prey. Ciliates ingest *Legionella* and protect this important pathogen from chlorine, which often is used in an attempt to control *Legionella* in cooling towers and air-conditioning units. The ciliate serves as a reservoir host. *Legionella pneumophila* also has been found to have a greater potential to invade macrophages and epithelial cells after predation, indicating that ingestion not only provides protection but also may make the bacterium a better pathogen. A similar phenomenon of survival in protozoa been observed for *Mycobac-*

Table 28.3 The Many Faces of Predation

Predation Result	Example
Digestion	The microbial loop. Soluble organic matter from primary producers is normally used by bacteria, which become a particulate food source for higher consumers. Flagellates and ciliates prey on these bacteria and digest them, making the nutrients they contain available again in mineral form for use in primary production, creating the microbial loop. In this way a large portion of the carbon fixed by the photosynthetic microbes is mineralized and recycled (thus the term microbial loop) and does not reach the higher trophic levels of the ecosystem (see figure 29.4). Predation also can reduce the density-dependent stress factors in prey populations, allowing more rapid growth and turnover of the prey than would occur if the predator were not active.
Retention	Bacteria retained within the predator serve a useful purpose, as in the transformation of toxic hydrogen produced by ciliates in the rumen to harmless methane. Also, trapping of chloroplasts (kleptochloroplasty) by protozoa provides the predator with photosynthate.
Protection and increased fitness	The intracellular survival of <i>Legionella</i> ingested by ciliates protects it from stresses such as heating and chlorination. Ingestion also results in increased pathogenicity when the prey is again released to the external environment, and this may be required for infection of humans. The predator serves as a reservoir host. Nanoplankton may be ingested by zooplankton and grow in the zooplankton digestive system. They are then released to the environment in a fitter state. Dissemination to new locations also occurs.

terium avium, a pathogen of worldwide concern. These protective aspects of predation have major implications for survival and control of disease-causing microorganisms in the biofilms present in water supplies and air-conditioning systems. In marine systems the ingestion of nanoplankton by zooplankton provides a nutrient-rich environment that allows nanoplankton reproduction in the digestive tract and promotes dissemination in the environment. A similar process occurs after bacteria are ingested by polychaetes (segmented worms found mostly in marine environments).

Fungi often show interesting predatory skills. Some fungi can trap protozoa by the use of sticky hyphae or knobs, sticky networks of hyphae, or constricting or nonconstricting rings. A classic example is *Arthrobotrys*, which traps nematodes by use of constricting rings. After the nematode is trapped, hyphae grow into the immobilized prey and the cytoplasm is used as a nutrient. Other fungi have conidia that, after ingestion by an unsuspecting prey, grow and attack the susceptible host from inside the intestinal tract. In this situation the fungus penetrates the host cells in a complex interactive process.

Thus predation, which usually has a fatal and final outcome for an individual prey organism, has a wide range of beneficial effects on prey populations, and is critical in the functioning of natural environments.

Parasitism

Parasitism is one of the most complex microbial interactions; the line between parasitism and predation is difficult to define (figure 28.1; see also section 34.1). This is a relationship in which one of a pair benefits from the other, and the host is usually harmed. This can involve nutrient acquisition and/or physical maintenance in or on the host. In parasitism there is a degree of coexistence of the parasite in association with the host. Depending on the equilibrium between the two organisms, this may shift and what might have been a stable parasitic relationship may then become a pathogenic one which can be defined as predation.

Some bacterial viruses can establish a lysogenic relationship with their hosts, and the viruses, in their prophage state, can confer positive new attributes on the host bacteria, as occurs with toxin production by *Corynebacterium diphtheriae* (see sections 17.5 and 34.3). Parasitic fungi include *Rhizophydium sphaerocarpum* with the alga *Spyrogyra*. Also, *Rhizoctonia solani* is a parasite of *Mucor* and *Pythium*, which is important in **biocontrol processes**, the use of one microorganism to control another. Human diseases caused by viruses, bacteria, fungi, and protozoa are discussed in chapters 38 through 40.

1. Define predation and parasitism. How are these similar and different?
2. How can a predator confer positive benefits on its prey? Think of the responses of individual organisms versus populations as you consider this question.
3. What are examples of parasites that are important in microbiology?

Amensalism

Amensalism (from the Latin for *not* at the same table) describes the negative effect that one organism has on another organism as shown in figure 28.1. This is a unidirectional process based on the release of a specific compound by one organism which has a negative effect on another organism. A classic example of amensalism is the production of antibiotics that can inhibit or kill a susceptible microorganism (figure 28.16a). The attine ant-fungal mutualistic relationship is promoted by antibiotic-producing bacteria that are maintained in the fungal garden system (figure 28.16b). In this case a streptomycete produces an antibiotic that controls *Escovopsis*, a persistent parasitic fungus that can destroy the ant's fungal garden. This unique amensalistic process appears to have evolved 50 million years ago in South America.

Other important amensalistic relationships involve microbial production of specific organic compounds that disrupt cell wall or plasma membrane integrity. These include the bacteriocins (see p. 297, 712). These substances are of increasing interest as food additives for controlling growth of undesired pathogens (see section 41.3). Antibacterial peptides can be released by the host and microorganisms in the intestine. These molecules, called cecropins in insects and defensins in mammals, recently have been recognized as effector molecules that play significant roles in innate immunity (see p. 720). In animals these molecules are released by phagocytes and intestinal cells, and are as powerful as tetracyclines. Finally, metabolic products, such as organic acids formed in fermentation, can produce amensalistic effects. These compounds inhibit growth by changing the environmental pH, for example, during natural milk spoilage (see section 41.2).

Competition

Competition arises when different microorganisms within a population or community try to acquire the same resource, whether this is a physical location or a particular limiting nutrient (figure 28.1). This principle of competition was studied by E. F. Gause, who in 1934 described this as the **competitive exclusion principle**. He found that if the two competing ciliates overlapped too much in terms of their resource use, one of the two protozoan populations was excluded. In chemostats (see section 6.3), we may see competition for a limiting nutrient among microorganisms with transport systems of differing affinity. This can lead to the exclusion of the slower-growing population under a particular set of conditions. If the dilution rate is changed, the previously slower-growing population may become dominant. Often two microbial populations that appear to be similar nevertheless coexist. In these cases there is a subtle difference in the characteristics of the microorganisms or their microenvironments that make this coexistence possible.

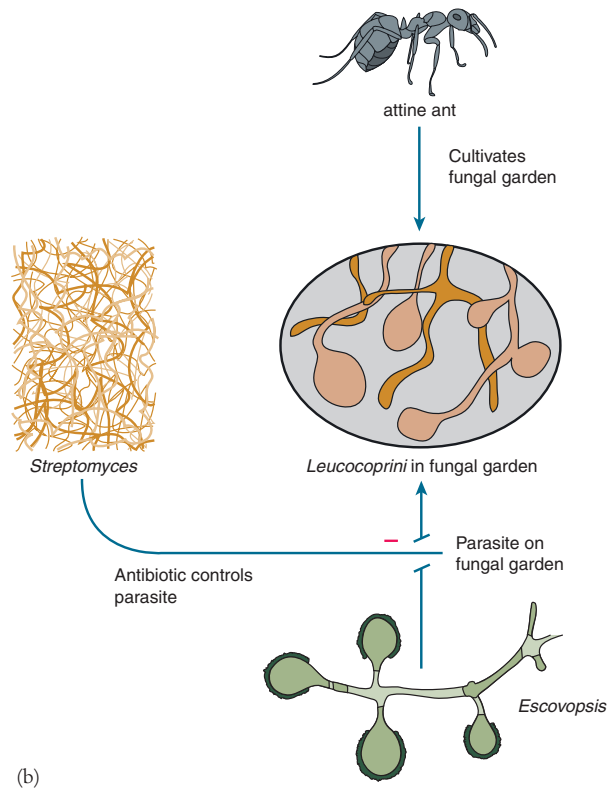
1. What is the origin of the term amensalism?
2. What are bacteriocins?
3. What is the competitive exclusion principle?



(a)

Figure 28.16 Amensalism: A Negative Microbe-Microbe Interaction.

Interaction. (a) Antibiotic production and inhibition of growth of a susceptible bacterium on an agar medium. (b) A schematic diagram describing the use of antibiotic-producing streptomycetes by ants to control fungal parasites in their fungal garden.



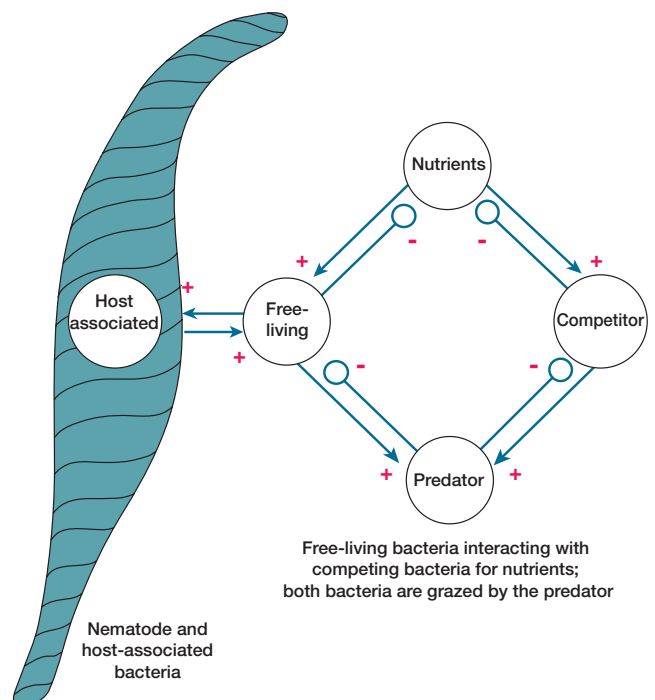
(b)

Symbioses in Complex Systems

It should be emphasized that the symbiotic interactions discussed in this section do not occur independently. Each time a microorganism interacts with other organisms and their environments, a series of feedback responses occurs in the larger biotic community that will impact other parts of ecosystems. As an illustration, the interactions between the nematode *Eubostrichus parasitiferus* and its protozoan sulfide-oxidizing bacterium (p. 606) involves a series of symbiotic interactions, as shown in figure 28.17. In this case, the protozoan interaction between the nematodes and the sulfide-oxidizing epibiont is influenced by the population size of the associated bacterium, and whether it is host-associated or free-living. This equilibrium between host-associated and free-

Figure 28.17 Complex Interactions in Microbial Ecology.

Interactions between the marine nematode *Eubostrichus parasitiferus* and its host-associated and free-living protozoan bacterium are influenced by levels of nutrients, as well as by other bacteria competing for these nutrients. Predators, in turn, use both the free-living protozoan bacterium and the competing bacteria as food sources, leading to complex feedback responses in this dynamic ecosystem. Arrows show positive or enhancing interactions; lines with small circles indicate negative or damping interactions.



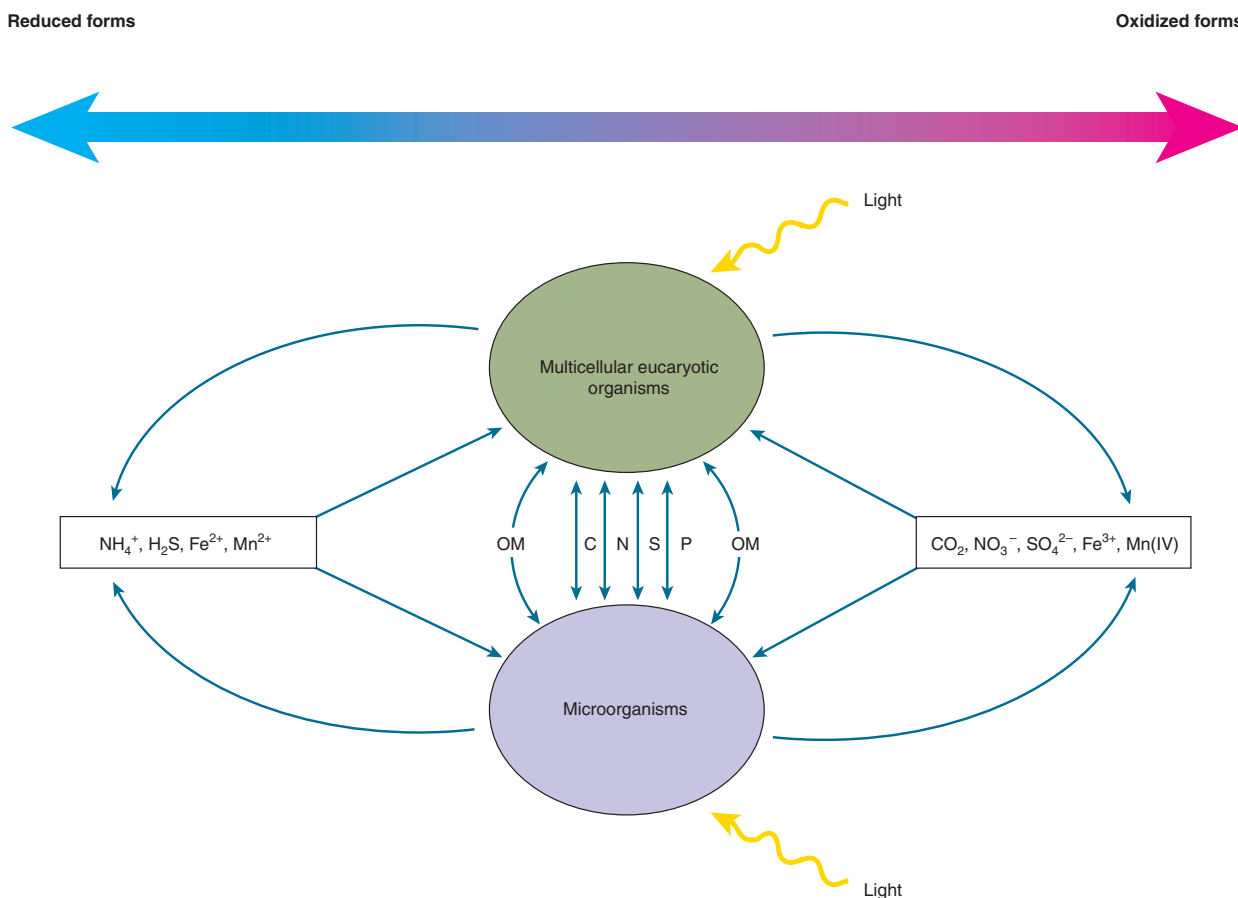


Figure 28.18 Macrobiogeochemistry: A Cosmic View of Mineral Cycling in Microorganisms, Higher Organisms, and the Abiotic Chemical World. All biogeochemical cycles are linked, with energy being obtained from light and pairs of reduced and oxidized compounds. Only major flows are shown. See individual cycles for details of energetic-linked relationships. The forms that move between the microorganisms and multicellular organisms can vary. The broad concept is that all cycles are linked. The biotic components include both living forms and those that have died/senesced and which are being processed. Flows from lithogenic sources are important for phosphorus. Methane transformations are only microbial (see discussion of the carbon cycle). Organic matter (OM).

living bacteria is controlled by a series of feedback processes that involve competition for sulfide with other bacteria, and predation on the epibiont and on the competing bacteria. Thus the equilibrium observed at any time is the result of a series of interactions, involving protooperation, predation, and competition.

28.3 Nutrient Cycling Interactions

Microorganisms, in the course of their growth and metabolism, interact with each other in the cycling of nutrients, including carbon, sulfur, nitrogen, phosphorus, iron, and manganese. This nutrient cycling, called **biogeochemical cycling** when applied to the environment, involves both biological and chemical processes. Nutrients are transformed and cycled, often by oxidation-reduction reactions (see section 8.5) that can change the chemical and

physical characteristics of the nutrients. All of the biogeochemical cycles are linked (**figure 28.18**), and the metabolism-related transformations of these nutrients have global-level impacts.

The major reduced and oxidized forms of the most important elements are noted in **table 28.4**, together with their valence states. Significant gaseous components occur in the carbon and nitrogen cycles and, to a lesser extent, in the sulfur cycles. Thus a soil, aquatic, or marine microorganism often can fix gaseous forms of carbon and nitrogen compounds. In the “sedimentary” cycles, such as that for iron, there is no gaseous component.

Carbon Cycle

Carbon can be present in reduced forms, such as methane (CH_4) and organic matter, and in more oxidized forms, such as carbon monoxide (CO) and carbon dioxide (CO_2). The major pools

Table 28.4 The Major Forms of Carbon, Nitrogen, Sulfur, and Iron Important in Biogeochemical Cycling

		Major Forms and Valences				
Cycle	Significant Gaseous Component Present?	Reduced Forms	Intermediate Oxidation State Forms			Oxidized Forms
C	Yes	CH ₄ (−4)	CO (+2)			CO ₂ (+4)
N	Yes	NH ₄ ⁺ , organic N (−3)	N ₂ (0)	N ₂ O (+1)	NO ₂ [−] (+3)	NO ₃ [−] (+5)
S	Yes	H ₂ S, SH groups in organic matter (−2)	S ⁰ (0)	S ₂ O ₃ ^{2−} (+2)	SO ₃ ^{2−} (+4)	SO ₄ ^{2−} (+6)
Fe	No	Fe ²⁺ (+2)				Fe ³⁺ (+3)

Note: The carbon, nitrogen, and sulfur cycles have significant gaseous components, and these are described as gaseous nutrient cycles. The iron cycle does not have a gaseous component, and this is described as a sedimentary nutrient cycle. Major reduced, intermediate oxidation state, and oxidized forms are noted, together with valences.

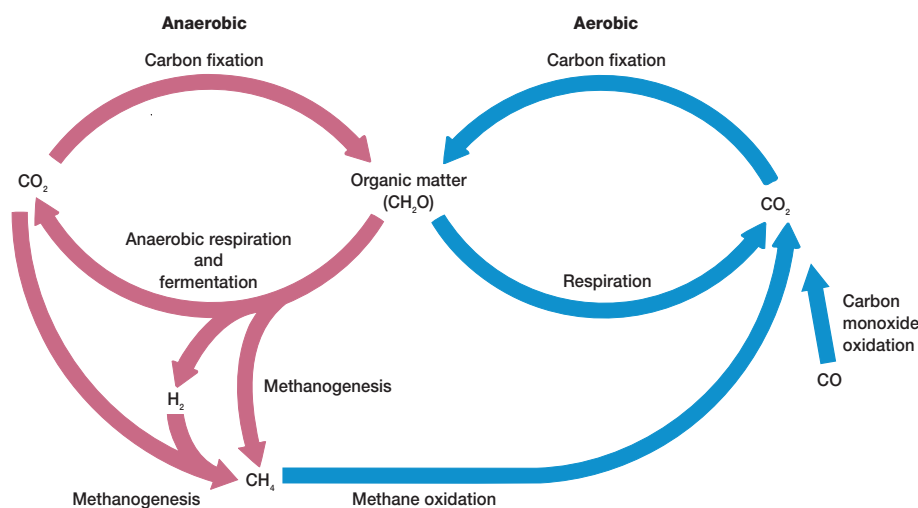


Figure 28.19 The Basic Carbon Cycle in the Environment. Carbon fixation can occur through the activities of photoautotrophic and chemoautotrophic microorganisms. Methane can be produced from inorganic substrates (CO₂ + H₂) or from organic matter. Carbon monoxide (CO)—produced by sources such as automobiles and industry—is returned to the carbon cycle by CO-oxidizing bacteria. Aerobic processes are noted with blue arrows, and anaerobic processes are shown with red arrows. Reverse methanogenesis will be discussed in chapter 29.

present in an integrated basic carbon cycle are shown in **figure 28.19**. Reductants (e.g., hydrogen, which is a strong reductant) and oxidants (e.g., O₂) influence the course of biological and chemical reactions involving carbon. Hydrogen can be produced during organic matter degradation especially under anaerobic conditions when fermentation occurs. If hydrogen and methane are generated, they can move upward from anaerobic to aerobic areas. This creates an opportunity for aerobic hydrogen and methane oxidizers to function.

Methane levels in the atmosphere have been increasing approximately 1% per year, from 0.7 to 1.6 to 1.7 ppm (volume) in the last

300 years. This methane is derived from a variety of sources. If an aerobic water column is above the anaerobic zone where the methanogens are located, the methane can be oxidized before it reaches the atmosphere. In many situations, such as in rice paddies without an overlying aerobic water zone, the methane will be released directly to the atmosphere, thus contributing to global atmospheric methane increases. Rice paddies, ruminants, coal mines, sewage treatment plants, landfills, and marshes are important sources of methane. Anaerobic microorganisms such as *Methanobrevibacter* in the guts of termites also can contribute to methane production.

Physiology of aerobic hydrogen and methane utilizers (pp. 193, 502–3)

Table 28.5 Complex Organic Substrate Characteristics That Influence Decomposition and Degradability

Substrate	Basic Subunit	Linkages (if Critical)	Elements Present in Large Quantity					Degradation	
			C	H	O	N	P	With O ₂	Without O ₂
Starch	Glucose	$\alpha(1 \rightarrow 4)$	+	+	+	-	-	+	+
		$\alpha(1 \rightarrow 6)$							
Cellulose	Glucose	$\beta(1 \rightarrow 4)$	+	+	+	-	-	+	+
Hemicellulose	C6 and C5 monosaccharides	$\beta(1 \rightarrow 4)$, $\beta(1 \rightarrow 3)$, $\beta(1 \rightarrow 6)$	+	+	+	-	-	+	+
Lignin	Phenylpropane	C-C, C-O bonds	+	+	+	-	-	+	-
Chitin	N-acetylglucosamine	$\beta(1 \rightarrow 4)$	+	+	+	+	-	+	+
Protein	Amino acids	Peptide bonds	+	+	+	+	-	+	+
Hydrocarbon	Aliphatic, cyclic, aromatic		+	+	-	-	-	+	+/-
Lipids	Glycerol, fatty acids; some contain phosphate and nitrogen	Esters	+	+	+	+	+	+	+
Microbial biomass		Varied	+	+	+	+	+	+	+
Nucleic acids	Purine and pyrimidine bases, sugars, phosphate	Phosphodiester and N-glycosidic bonds	+	+	+	+	+	+	+

Carbon fixation occurs through the activities of cyanobacteria and green algae, photosynthetic bacteria (e.g., *Chromatium* and *Chlorobium*), and aerobic chemolithoautotrophs.

In the carbon cycle depicted in figure 28.19, no distinction is made between different types of organic matter that are formed and degraded. This is a marked oversimplification because organic matter varies widely in physical characteristics and in the biochemistry of its synthesis and degradation. Organic matter varies in terms of elemental composition, structure of basic repeating units, linkages between repeating units, and physical and chemical characteristics.

The formation of organic matter is discussed in chapters 10 through 12. The degradation of this organic matter, once formed, is influenced by a series of factors. These include (1) nutrients present in the environment; (2) abiotic conditions (pH, oxidation-reduction potential, O₂, osmotic conditions), and (3) the microbial community present.

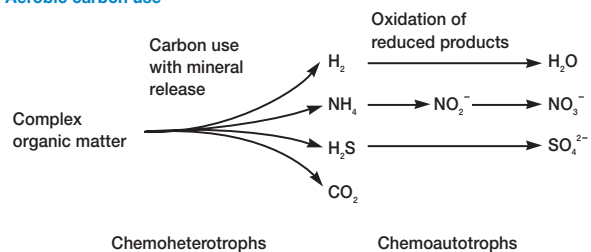
The major complex organic substrates used by microorganisms are summarized in **table 28.5**. Of these, only previously grown microbial biomass contains all of the nutrients required for microbial growth. Chitin, protein, microbial biomass, and nucleic acids contain nitrogen in large amounts. If these substrates are used for growth, the excess nitrogen and other minerals that are not used in the formation of new microbial biomass will be released to the environment, in the process of **mineralization**. This is the process in which organic matter is decomposed to release simpler, inorganic compounds (e.g., CO₂, NH₄⁺, CH₄, H₂).

The other complex substrates in table 28.5 contain only carbon, hydrogen, and oxygen. If microorganisms are to grow by using these substrates, they must acquire the remaining nutrients they need for biomass synthesis from the environment; in the process of **immobilization**.

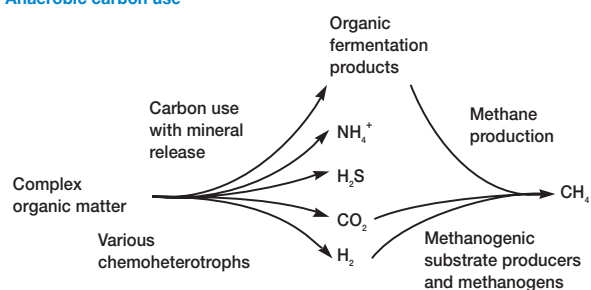
The oxygen relationships for the use of these substrates also are of interest, because most of them can be degraded easily with or without oxygen present. The exceptions are hydrocarbons and lignin. Hydrocarbons are unique in that microbial degradation, especially of straight-chained and branched forms, involves the initial addition of molecular O₂. Recently, anaerobic degradation of hydrocarbons with sulfate or nitrate as oxidants has been observed. With sulfate present, organisms of the genus *Desulfovibrio* are active. This occurs only slowly and with microbial communities that have been exposed to these compounds for extended periods. Such degradation may have resulted in the sulfides that are present in "sour gases" associated with petroleum.

Lignin, an important structural component in mature plant materials, is a complex amorphous polymer based on a phenylpropane building block, linked by carbon-carbon and carbon-ether bonds. It makes up approximately 1/3 of the weight of wood. This is a special case in which biodegradability is dependent on O₂ availability. There often is no significant degradation because most filamentous fungi that degrade native lignin in situ can function only under aerobic conditions where oxidases can act by the release of active oxygen species. Lignin's lack of biodegradability under anaerobic conditions results in accumulation of lignified materials, including the formation of peat bogs and muck soils. This absence of lignin degradation under anaerobic conditions also is important in construction. Large masonry structures often are built on swampy sites by driving in wood pilings below the water table and placing the building footings on the pilings. As long as the foundations remain water-saturated and anaerobic, the structure is stable. If the water table drops, however, the pilings will begin to rot and the structure will be threatened. Similarly, the cleanup of harbors can lead to decomposition of costly docks built with wooden pilings due to increased aerobic degradation of wood by filamentous fungi. Rumen function provides a final example of the relationship between

Aerobic carbon use



Anaerobic carbon use

**Figure 28.20 The Influence of Oxygen on Organic Matter**

Decomposition. Microorganisms form different products when breaking down complex organic matter aerobically than they do under anaerobic conditions. Under aerobic conditions oxidized products accumulate, while reduced products accumulate anaerobically. These reactions also illustrate commensalistic transformations of a substrate, where the waste products of one group of microorganisms can be used by a second type of microorganism.

lignin degradation and oxygen. The rumen (pp. 602–4), being almost free of oxygen, does not allow significant degradation of lignin present in animal feeds. The use of sugars and carbohydrates in the rumen leaves an inactive residue that can improve soils more effectively than the original feeds.

Patterns of microbial degradation are important in many habitats. They contribute to the accumulation of petroleum products, the formation of bogs, and the preservation of valuable historical objects.

The presence or absence of oxygen also affects the final products that accumulate when organic substrates have been processed by microorganisms and mineralized either under aerobic or anaerobic conditions. Under aerobic conditions, oxidized products such as nitrate, sulfate, and carbon dioxide (**figure 28.20**) will result from microbial degradation of complex organic matter. In comparison, under anaerobic conditions reduced end products tend to accumulate, including ammonium ion, sulfide, and methane.

These oxidized and reduced forms, if they remain in the aerobic or anaerobic environments where they were formed, will usually only serve as nutrients. If mixing occurs, oxidized species might be moved to a more reduced zone or reduced species might be moved to a more oxidized zone. Under such circumstances, additional energetic possibilities (linking of oxidants and reductants) will be created, leading to succession and further nutrient

cycling as these mixed oxidants and reductants are exploited by the microbial community.

1. What is biogeochemical cycling?
2. Which organic polymers discussed in this section do and do not contain nitrogen?
3. What is unique about lignin and its degradation?
4. Define mineralization and immobilization and give examples.
5. What C, N, and S forms will accumulate after anaerobic degradation of organic matter?

Sulfur Cycle

Microorganisms contribute greatly to the sulfur cycle, a simplified version of which is shown in **figure 28.21**. Photosynthetic microorganisms transform sulfur by using sulfide as an electron source, allowing *Thiobacillus* and similar chemolithoautotrophic genera to function (see pp. 193–94 and 496–98). In contrast, when sulfate diffuses into reduced habitats, it provides an opportunity for different groups of microorganisms to carry out **sulfate reduction**. For example, when a usable organic reductant is present, *Desulfovibrio* uses sulfate as an oxidant (see pp. 190 and 507–10). This use of sulfate as an external electron acceptor to form sulfide, which accumulates in the environment, is an example of a **dissimilatory reduction** process and anaerobic respiration. In comparison, the reduction of sulfate for use in amino acid and protein biosynthesis is described as an **assimilatory reduction** process (see section 10.4). Other microorganisms have been found to carry out dissimilatory elemental sulfur reduction. These include *Desulfuromonas* (see pp. 507–10), thermophilic archaea (see chapter 20), and also cyanobacteria in hypersaline sediments. Sulfite is another critical intermediate that can be reduced to sulfide by a wide variety of microorganisms, including *Alteromonas* and *Clostridium*, as well as *Desulfovibrio* and *Desulfotomaculum*. *Desulfovibrio* is usually considered as an obligate anaerobe. Recent research, however, has shown that this interesting organism also respire using oxygen, when it is present at a dissolved oxygen level of 0.04%.

In addition to the very important photolithotrophic sulfur oxidizers such as *Chromatium* and *Chlorobium*, which function under strict anaerobic conditions in deep water columns, a large and varied group of bacteria carry out **aerobic anoxygenic photosynthesis**. These aerobic anoxygenic phototrophs use bacteriochlorophyll *a* and carotenoid pigments and are found in marine and freshwater environments; they are often components of microbial mat communities. Important genera include *Erythronomas*, *Roseococcus*, *Porphyrobacter*, and *Roseobacter*.

“Minor” compounds in the sulfur cycle play major roles in biology. An excellent example is dimethylsulfoniopropionate (DMSP), which is used by bacterioplankton (floating bacteria) as a sulfur source for protein synthesis, and which is transformed to dimethylsulfide (DMS), a volatile sulfur form that can affect atmospheric processes.

When pH and oxidation-reduction conditions are favorable, several key transformations in the sulfur cycle also occur as the

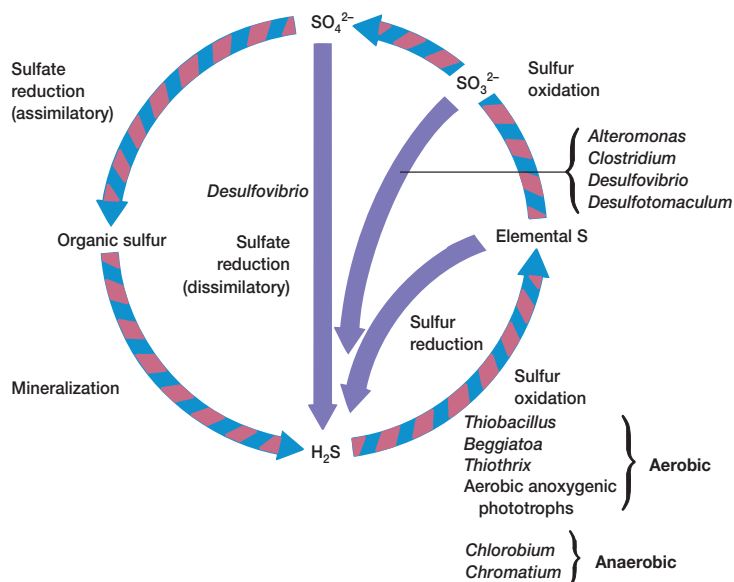


Figure 28.21 The Basic Sulfur Cycle. Photosynthetic and chemosynthetic microorganisms contribute to the environmental sulfur cycle. Sulfate and sulfite reductions carried out by *Desulfovibrio* and related microorganisms, noted with purple arrows, are dissimilatory processes. Sulfate reduction also can occur in assimilatory reactions, resulting in organic sulfur forms. Elemental sulfur reduction to sulfide is carried out by *Desulfuromonas*, thermophilic archaea, or cyanobacteria in hypersaline sediments. Sulfur oxidation can be carried out by a wide range of aerobic chemotrophs and by aerobic and anaerobic phototrophs.

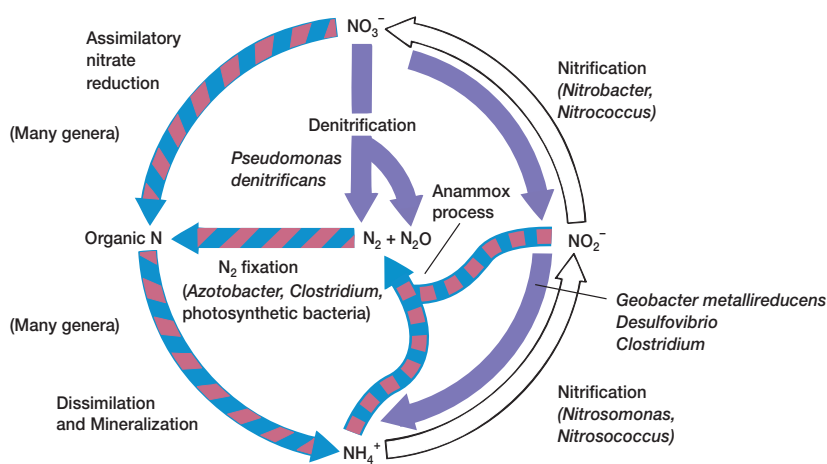


Figure 28.22 The Basic Nitrogen Cycle. Flows that occur predominantly under aerobic conditions are noted with open arrows. Anaerobic processes are noted with solid bold arrows. Processes occurring under both aerobic and anaerobic conditions are marked with cross-barred arrows. The anammox reaction of NO_2^- and NH_4^+ to yield N_2 is shown. Important genera contributing to the nitrogen cycle are given as examples.

result of chemical reactions in the absence of microorganisms. An important example of such an abiotic process is the oxidation of sulfide to elemental sulfur. This takes place rapidly at a neutral pH, with a half-life of approximately 10 minutes for sulfide at room temperature.

Nitrogen Cycle

Several important aspects of the basic nitrogen cycle will be discussed: the processes of nitrification, denitrification, and nitrogen fixation (**figure 28.22**). It should be emphasized that this is a “basic” nitrogen cycle. Although not mentioned in the figure, the heterotrophs can carry out nitrification, and some of these heterotrophs combine nitrification with anaerobic denitrification, thus oxidizing ammonium ion to N_2O and N_2 with depressed oxy-

gen levels. The occurrence of anoxic ammonium ion oxidation (anammox is the term used for the commercial process) means that nitrification is not only an aerobic process. Thus as we learn more about the biogeochemical cycles, including that of nitrogen, the simple cycles of earlier textbooks are no longer accurate representations of biogeochemical processes.

Nitrification is the aerobic process of ammonium ion (NH_4^+) oxidation to nitrite (NO_2^-) and subsequent nitrite oxidation to nitrate (NO_3^-). Bacteria of the genera *Nitrosomonas* and *Nitrosococcus*, for example, play important roles in the first step, and *Nitrobacter* and related chemolithoautotrophic bacteria carry out the second step. Recently *Nitrosomonas eutropha* has been found to oxidize ammonium ion anaerobically to nitrite and nitric oxide (NO) using nitrogen dioxide (NO_2) as an oxidant in a denitrification-related reaction. In addition, **heterotrophic nitrification** by

bacteria and fungi contributes significantly to these processes in more acidic environments. [Nitrification and nitrifiers \(p. 193\)](#)

The process of **denitrification** requires a different set of environmental conditions. This dissimilatory process, in which nitrate is used as an oxidant in anaerobic respiration, usually involves heterotrophs such as *Pseudomonas denitrificans*. The major products of denitrification include nitrogen gas (N_2) and nitrous oxide (N_2O), although nitrite (NO_2^-) also can accumulate. Nitrite is of environmental concern because it can contribute to the formation of carcinogenic nitrosamines. Finally, nitrate can be transformed to ammonia in dissimilatory reduction by a variety of bacteria, including *Geobacter metallireducens*, *Desulfovibrio* spp., and *Clostridium*. [Denitrification and anaerobic respiration \(pp. 190–91\)](#)

Nitrogen assimilation occurs when inorganic nitrogen is used as a nutrient and incorporated into new microbial biomass. Ammonium ion, because it is already reduced, can be directly incorporated without major energy costs. However, when nitrate is assimilated, it must be reduced with a significant energy expenditure. In this process nitrite may accumulate as a transient intermediate. [The biochemistry of nitrogen assimilation \(pp. 210–14\)](#)

Nitrogen fixation can be carried out by aerobic or anaerobic procaryotes and does not occur in eucaryotes. Under aerobic conditions a wide range of free-living microbial genera (*Azotobacter*, *Azospirillum*) contribute to this process. Under anaerobic conditions the most important free-living nitrogen fixers are members of the genus *Clostridium*. Nitrogen fixation by cyanobacteria such as *Anabaena* and *Oscillatoria* can lead to the enrichment of aquatic environments with nitrogen. These nutrient-enrichment processes are discussed in chapter 29. In addition, nitrogen fixation can occur through the activities of bacteria that develop symbiotic associations with plants. These associations include *Rhizobium* and *Bradyrhizobium* with legumes, *Frankia* in association with many woody shrubs, and *Anabaena*, with *Azolla*, a water fern important in rice cultivation. [The establishment of the *Rhizobium*-legume association \(pp. 675–78\)](#)

The nitrogen-fixation process involves a sequence of reduction steps that require major energy expenditures. Ammonia, the product of nitrogen reduction, is immediately incorporated into organic matter as an amine. Reductive processes are extremely sensitive to O_2 and must occur under anaerobic conditions even in aerobic microorganisms. Protection of the nitrogen-fixing enzyme is achieved by means of a variety of mechanisms, including physical barriers, as occurs with heterocysts in some cyanobacteria (*see section 21.3*), O_2 scavenging molecules, and high rates of metabolic activity. [The biochemistry of nitrogen fixation \(pp. 212–14\)](#)

As shown in figure 28.22, microorganisms have been isolated that can couple the anaerobic oxidation of NH_4^+ with the reduction of NO_2^- , to produce gaseous nitrogen, in what has been termed the **anammox process** (*anoxic ammonia oxidation*). This may provide a means by which nitrogen can be removed from sewage plant effluents to decrease nitrogen flow to sensitive freshwater and marine ecosystems. It has been suggested that chemolithotrophic members of the planctomycetes (*see section 21.4*) play a role in this process.

1. What are the major oxidized and reduced forms of sulfur and nitrogen?
2. Diagram a simple sulfur cycle.
3. Why is dimethylsulfide (DMS), considered to be a “minor” part of the sulfur cycle, of such environmental importance?
4. What is aerobic anoxygenic photosynthesis?
5. What are nitrification, denitrification, nitrogen fixation, and the anammox process?

Iron Cycle

The iron cycle (figure 28.23) includes several different genera that carry out iron oxidations, transforming ferrous ion (Fe^{2+}) to ferric ion (Fe^{3+}). *Thiobacillus ferrooxidans* carries out this process under acidic conditions, *Gallionella* is active under neutral pH conditions, and *Sulfolobus* functions under acidic, thermophilic conditions. Much of the earlier literature suggested that additional genera could oxidize iron, including *Sphaerotilus* and *Leptothrix*. These two genera are still termed “iron bacteria” by many nonmicrobiologists. Confusion about the role of these genera resulted from the occurrence of the chemical oxidation of ferrous ion to ferric ion (forming insoluble iron precipitates) at neutral pH values, where microorganisms also grow on organic substrates. These microorganisms are now classified as chemoheterotrophs.

Recently microbes have been found that oxidize Fe^{2+} using nitrate as an electron acceptor. This process occurs in aquatic sediments with depressed levels of oxygen and may be another route by which large zones of oxidized iron have accumulated in environments with lower oxygen levels.

Iron reduction occurs under anaerobic conditions resulting in the accumulation of ferrous ion. Although many microorganisms can reduce small amounts of iron during their metabolism, most iron reduction is carried out by specialized iron-respiring microorganisms such as *Geobacter metallireducens*, *Geobacter sulfurreducens*, *Ferribacterium limneticum*, and *Shewanella putrefaciens*, which can obtain energy for growth on organic matter using ferric iron as an oxidant.

In addition to these relatively simple reductions to ferrous ion, some magnetotactic bacteria such as *Aquaspirillum magnetotacticum* (*see section 3.3*) transform extracellular iron to the mixed valence iron oxide mineral magnetite (Fe_3O_4) and construct intracellular magnetic compasses. Furthermore, dissimilatory iron-reducing bacteria accumulate magnetite as an extracellular product.

Magnetite has been detected in sediments, where it is present in particles similar to those found in bacteria, indicating a longer-term contribution of bacteria to iron cycling processes. Genes for magnetite synthesis have been cloned into other organisms, creating new magnetically sensitive microorganisms. Magnetotactic bacteria are now described as **magneto-aerotactic bacteria**, due to their using magnetic fields to migrate to the position in a bog or swamp where the oxygen level is best suited for their functioning. In the last decade new microorganisms have been discovered that use ferrous ion as an electron donor in anoxygenic photosynthesis. Thus, with production of ferric ion in lighted

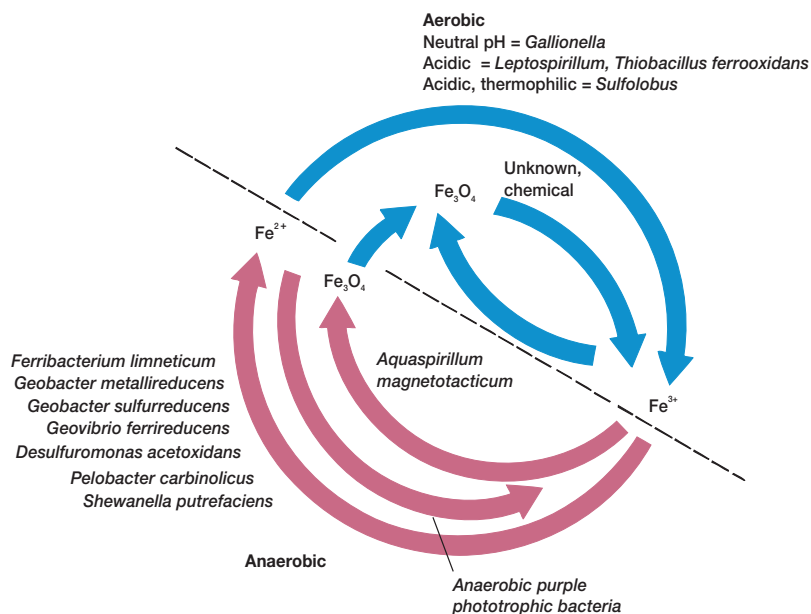


Figure 28.23 The Basic Iron Cycle. A simplified iron cycle with examples of microorganisms contributing to these oxidation and reduction processes. In addition to ferrous ion (Fe^{2+}) oxidation and ferric ion (Fe^{3+}) reduction, magnetite (Fe_3O_4), a mixed valence iron compound formed by magnetotactic bacteria is important in the iron cycle. Different microbial groups carry out the oxidation of ferrous ion depending on environmental conditions.

anaerobic zones by iron-oxidizing bacteria, the stage is set for subsequent chemotrophic-based iron reduction, such as by *Geobacter* and *Shewanella*, creating a strictly anaerobic oxidation/reduction cycle for iron.

Manganese Cycle

The importance of microorganisms in manganese cycling is becoming much better appreciated. The manganese cycle (figure 28.24) involves the transformation of manganous ion (Mn^{2+}) to MnO_2 (equivalent to manganic ion [Mn^{4+}]), which occurs in hydrothermal vents, bogs, and as an important part of rock varnishes. *Leptothrix*, *Arthrobacter*, *Pedomicrobium*, and the incompletely characterized “Metallogenium” are important in Mn^{2+} oxidation. *Shewanella*, *Geobacter*, and other chemoorganotrophs can carry out the complementary manganese reduction process.

Other Cycles and Cycle Links

Microorganisms can use a wide variety of additional metals as electron acceptors. Metals such as europium, tellurium, selenium, and rhodium can be reduced. Important microorganisms that reduce these metals include the photoorganotrophs *Rhodobacter*, *Rhodospirillum* and *Rhodopseudomonas*. For selenium, *Pseudomonas stutzeri*, *Thauera selenatis*, and *Wolinella succinogenes* are active. Such reductions can decrease the toxicity of a metal.

The microbial transformation of phosphorus involves primarily the transformation of phosphorus (+5 valence) from simple orthophosphate to various more complex forms, including polyphosphates found in metachromatic granules (see p. 52). A unique (and possibly microbial) product is phosphine (PH_3) with a −3 valence, which is liberated from swamps, soils, and marine regions and which ignites when exposed to air. This can then ignite methane

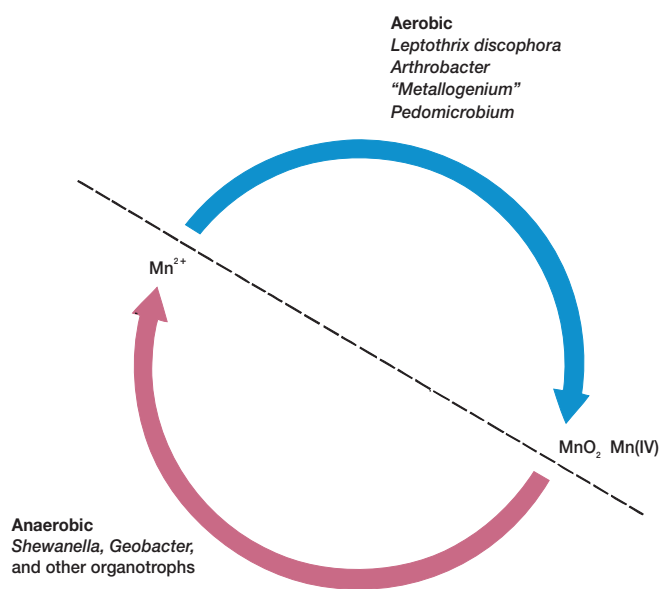


Figure 28.24 The Basic Manganese Cycle. Microorganisms make important contributions to the manganese cycle. Manganous ion (2+) is oxidized to manganic oxide (valence equivalent to 4+). Manganous oxide reduction is noted with a maroon arrow. Examples of organisms carrying out these processes are given.

produced in the same environment! The production of methane by anaerobic microorganisms will be discussed in the next chapter.

Having described the sulfur, iron, and manganese cycles as functioning independently, it is important to again emphasize that many microorganisms metabolically link these cycles by using

Table 28.6 Examples of Microorganism-Metal Interactions and Relations to Effects on Microorganisms and Homeothermic Animals

Interactions and Transformations				
Metal Group	Metal		Microorganisms	Homeothermic Animals
Noble metals	Ag	Silver	Microorganisms can reduce ionic forms to the elemental state. Low levels of ionized metals released to the environment have antimicrobial activity.	Many of these metals can be reduced to elemental forms and do not tend to cross the blood-brain barrier. Silver reduction can lead to inert deposits in the skin.
	Au	Gold		
	Pt	Platinum		
Metals that form stable carbon metal bonds	As	Arsenic	Microorganisms can transform inorganic and organic forms to methylated forms, some of which tend to bioaccumulate in higher trophic levels.	Methylated forms of some metals can cross the blood-brain barrier, resulting in neurological effects or death.
	Hg	Mercury		
	Se	Selenium		
Other metals	Cu	Copper	In the ionized form, at higher concentrations, these metals can directly inhibit microorganisms. They are often required at lower concentrations as trace elements.	At higher levels, clearance from higher organisms occurs by reaction with plasma proteins and other mechanisms. Many of these metals serve as trace elements at lower concentrations.
	Zn	Zinc		
	Co	Cobalt		

commonly shared oxidants and reductants. For example, some sulfate-reducing microorganisms can reduce Fe^{3+} using H_2 or organic matter as a reductant and also can oxidize elemental sulfur to sulfate when Mn(IV) is present as an electron acceptor. The Mn(IV)-dependent production of sulfate under anaerobic conditions, carried out by *Desulfobulbus propionicus*, provides a way to link sulfur and manganese cycling anaerobically.

Recently, a unique energetic couple has been described by B. Schinck and M. Friedrich. They isolated a lithoautotrophic bacterium from anaerobic sediments that can link the oxidation of phosphite (PO_3^{3-}) to phosphate (PO_4^{3-}) with the reduction of sulfate to hydrogen sulfide. Schinck and Friedrich have suggested that this energetic cycle could have operated on the early Earth.

1. What major forms of iron, manganese, and phosphorus are important in biogeochemical cycling?
2. Why is *Aquaspirillum* considered to be a magneto-aerotactic bacterium?
3. What are some important microbial genera that contribute to manganese cycling?
4. What is phosphine? Under which conditions will it be produced?
5. Describe some links between oxidants and reductants that have been discovered recently.

Microorganisms and Metal Toxicity

In addition to metals such as iron and manganese, which are largely nontoxic to microorganisms and animals, there are a series of metals that have varied toxic effects on microorganisms and homeothermic animals. Microorganisms play important roles in modifying the toxicity of these metals (table 28.6).

The “metals” can be considered in broad categories. The “noble metals” tend not to cross the vertebrate blood-brain barrier but can have distinct effects on microorganisms. Microorganisms also can reduce ionic forms of noble metals to their elemental forms.

The second group includes metals or metalloids that microorganisms can methylate to form more mobile products called organometals. Some organometals can cross the blood-brain barrier and affect the central nervous system of vertebrates. Organometals contain carbon-metal bonds. These bonds are their unique identifying characteristics.

The mercury cycle is of particular interest and illustrates many characteristics of those metals that can be methylated. Mercury compounds were widely used in industrial processes over the centuries. One has only to think of Lewis Carroll’s allusion to this problem when he wrote of the Mad Hatter in *Alice in Wonderland*. At that time mercury was used in the shaping of felt hats. Microorganisms methylated some of the mercury, thus rendering it more toxic to the hatmakers.

A devastating situation developed in southwestern Japan when large-scale mercury poisoning occurred in the Minamata Bay region because of industrial mercury released into the marine environment. Inorganic mercury that accumulated in bottom muds of the bay was methylated by anaerobic bacteria of the genus *Desulfovibrio* (figure 28.25). Such methylated mercury forms are volatile and lipid soluble, and the mercury concentrations increased in the food chain (by the process of **biomagnification**). The mercury was ultimately ingested by the human population, the “top consumers,” through their primary food source—fish—leading to severe neurological disorders.

A similar situation has occurred in many of the freshwater lakes in the north-central United States and in Canada, where mercury compounds were used to control microbial growth in pulp mills. Even decades later the fish in lakes downstream from these pulp mills cannot yet be used for food, and fishing is only for recreation.

The third group of metals occurs in ionic forms directly toxic to microorganisms. The metals in this group also can affect more complex organisms. However, plasma proteins react with the ionic forms of these metals and aid in their excretion unless excessive long-term contact and ingestion occur. Relatively high doses of these metals are required to cause lethal effects. At lower concentrations many of these metals serve as required trace elements.

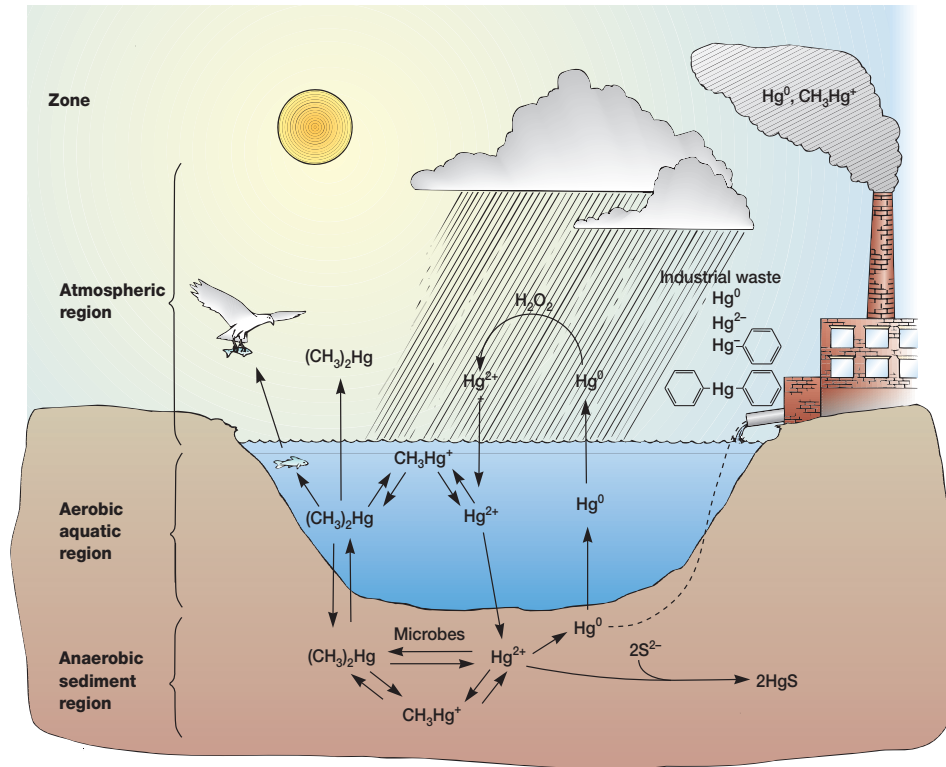


Figure 28.25 The Mercury Cycle. Interactions between the atmosphere, aerobic water, and anaerobic sediment are critical. Microorganisms in anaerobic sediments, primarily *Desulfovibrio*, can transform mercury to methylated forms that can be transported to water and the atmosphere. These methylated forms also undergo biomagnification. The production of volatile elemental mercury (Hg^0) releases this metal to waters and the atmosphere. Sulfide, if present in the anaerobic sediment, can react with ionic mercury to produce less soluble HgS .

The differing sensitivity of more complex organisms and microorganisms to metals forms the basis of many antiseptic procedures developed over the last 150 years (see section 7.5). The noble metals, although microorganisms tend to develop resistance to them, continue to be used in preference to antibiotics in some medical applications. Examples include the treatment of burns with silver-containing antimicrobial compounds and the use of silver-plated catheters.

1. What are examples of the three groups of metals in terms of their toxicity to microorganisms and homeothermic animals?
2. How can microbial activity render some metals more or less toxic to warm-blooded animals?
3. Why do metals such as mercury have such major effects on higher organisms?

28.4 The Physical Environment

Microorganisms, as they interact with each other and with other organisms in biogeochemical cycling, also are influenced by their immediate physical environment, whether this might be soil, water, the

deep marine environment, or a plant or animal host. It is important to consider the specific environments where microorganisms interact with each other, other organisms, and the physical environment.

The Microenvironment and Niche

The specific physical location of a microorganism is its **microenvironment**. In this physical microenvironment, the flux of required oxidants, reductants, and nutrients to the actual location of the microorganism can be limited. At the same time, waste products may not be able to diffuse away from the microorganism at rates sufficient to avoid growth inhibition by high waste product concentrations. These fluxes and gradients create a unique **niche**, which includes the microorganism, its physical habitat, the time of resource use, and the resources available for microbial growth and function (figure 28.26).

This physically structured environment also can limit the predatory activities of protozoa. If the microenvironment has pores with diameters of 3 to 6 μm , it will protect bacteria in the pores from predation, while allowing diffusion of nutrients and waste products. If the pores are larger, perhaps greater than 6 μm in diameter, protozoa may be able to feed on the bacteria. It is important to emphasize that microorganisms can create their own microenvironments and niches.

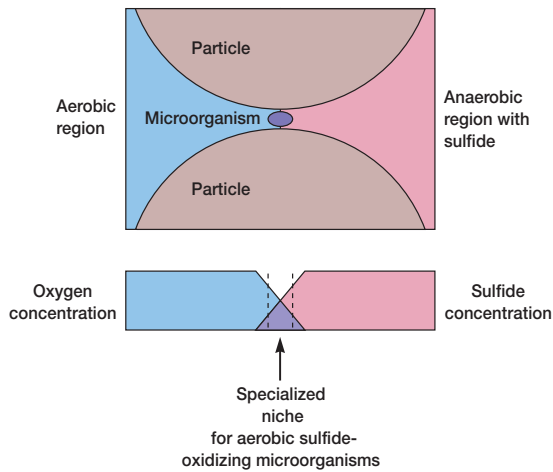


Figure 28.26 The Creation of a Niche from a Microenvironment.

As shown in this illustration, two nearby particles create a physical microenvironment for possible use by microorganisms. Chemical gradients, as with oxygen from the aerobic region, and sulfide from the anaerobic region, create a unique niche. This niche thus is the physical environment and the resources available for use by specialized aerobic sulfide-oxidizing bacteria.

For example, microorganisms in the interior of a colony have markedly different microenvironments and niches than those of the same microbial populations located on the surface or edge of the colony. Microorganisms also can associate with clays and form “clay hutches” for protection (*see section 42.4*).

1. What are the similarities and differences between a microenvironment and a niche?
2. Why might pores in soils, waters, and animals be important for survival of bacteria if protozoa are present?
3. Why might conditions vary for a bacterium on the edge of a colony in comparison with the center of the colony?

Biofilms and Microbial Mats

As noted in the previous section, microorganisms tend to create their own microenvironments and niches, even without having a structured physical environment available, by creating **biofilms**. These are organized microbial systems consisting of layers of microbial cells associated with surfaces. Such biofilms are an important factor in almost all areas of microbiology, as shown in **figure 28.27a**. Simple biofilms develop when microorganisms attach and form a monolayer of cells.

Depending on the particular microbial growth environment (light, nutrients present and diffusion rates), these biofilms can become more complex with layers of organisms of different types (*figure 28.27b*). A typical example would involve photosynthetic organisms on the surface, facultative chemoorganotrophs in the middle, and possibly sulfate-reducing microorganisms on the bottom.

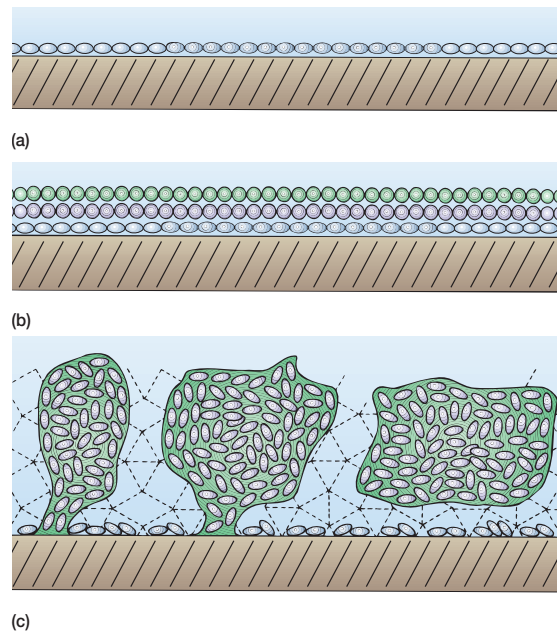


Figure 28.27 The Growth of Biofilms. Biofilms, or microbial growths on surfaces such as in freshwater and marine environments, can develop and become extremely complex, depending on the energy sources that are available. (a) Initial colonization by a single type of bacterium. (b) Development of a more complex biofilm with layered microorganisms of different types. (c) A mature biofilm with cell aggregates, interstitial pores, and conduits.

More complex biofilms can develop to form a four-dimensional structure (X, Y, Z, and time) with cell aggregates, interstitial pores, and conduit channels (*figure 28.27c*). This developmental process involves the growth of attached microorganisms, resulting in accumulation of additional cells on the surface, together with the continuous trapping and immobilization of free-floating microorganisms that move over the expanding biofilm. This structure allows nutrients to reach the biomass, and the channels are shaped by protozoa that graze on bacteria.

These more complex biofilms, in which microorganisms create unique environments, can be observed by the use of confocal scanning laser microscopy (CSLM) as discussed in chapter 2. The diversity of nonliving and living surfaces that can be exploited by biofilm-forming microorganisms is illustrated in **figure 28.28**. These include surfaces in catheters and dialysis units, which have intimate contact with human body fluids. Control of such microorganisms and their establishment in these sensitive medical devices is an important part of modern hospital care.

Microorganisms that form biofilms on living organisms such as plants or animals have additional advantages. In these cases the surfaces themselves often release nutrients, in the form of sloughed cells, soluble materials, and gases. These biofilms also can play major roles in disease (*see Box 39.3*) because they can protect pathogens from disinfectants, create a focus for later occurrence of

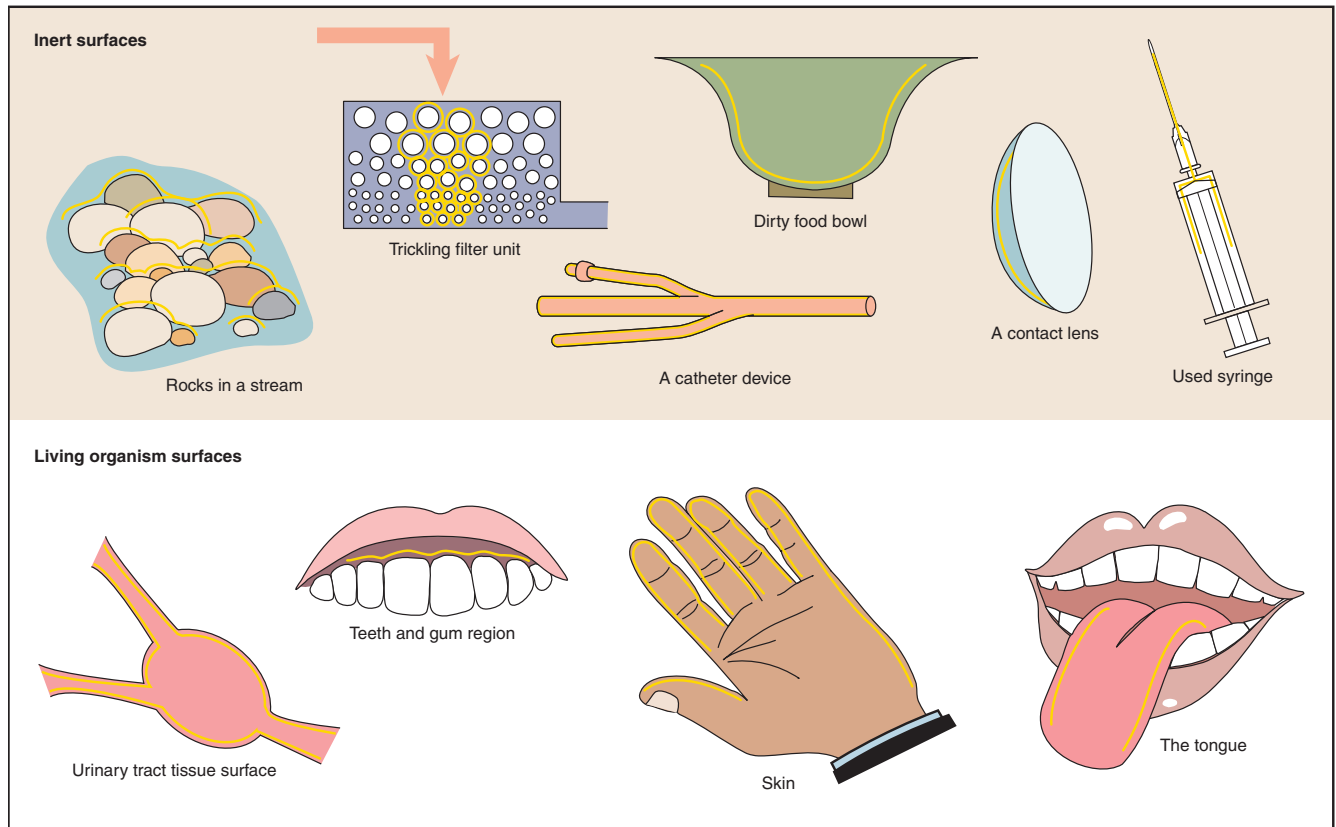


Figure 28.28 Biofilm Formation on Inert and Living Organism Surfaces. Biofilms, noted in yellow, are a part of microbial functioning in the environment, in biotechnology, and in human health.

disease, or release microorganisms and microbial products that may affect the immunological system of a susceptible host. Biofilms are critical in ocular diseases because *Chlamydia*, *Staphylococcus*, and other pathogens survive in ocular devices such as contact lenses and in cleaning solutions (**figure 28.29**).

Depending on environmental conditions, biofilms can become so large that they are visible and have macroscopic dimensions. Bands of microorganisms of different colors are evident as shown in **figure 28.30**. These thick biofilms, called **microbial mats**, are found in many freshwater and marine environments. These mats are complex layered microbial communities that can form at the surface of rocks or sediments in hypersaline and freshwater lakes, lagoons, hot springs, and beach areas. They consist of microbial filaments, including cyanobacteria. A major characteristic of mats is the extreme gradients that are present. Light only penetrates approximately 1 mm into these communities, and below this photosynthetic zone, anaerobic conditions occur and sulfate-reducing bacteria play a major role. The sulfide that these organisms produce diffuses to the anaerobic lighted region, allowing sulfur-dependent photosynthetic microorganisms to grow. Some believe that microbial mats could have allowed the

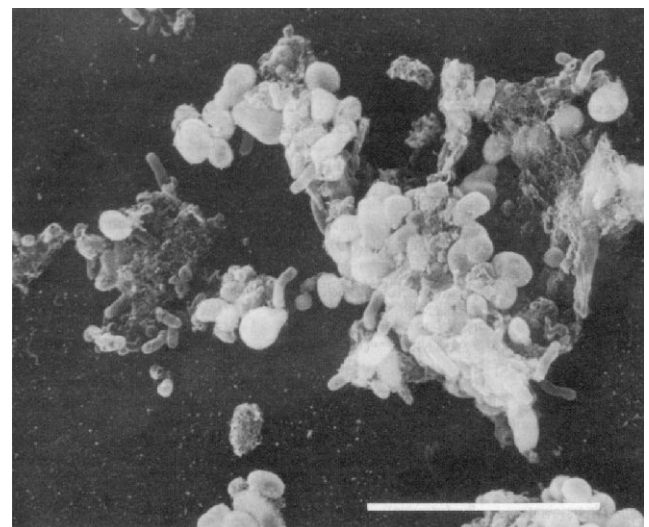


Figure 28.29 Contact Lenses Can Have Extensive Biofilms. Clumps of cocci and sparse rods on a contact lens. Bar = 10 μm .